



Commercial Drone Operation Training and Certification Requirements

Australia • Brazil • Canada • European Union • India
Israel • Japan

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Comparative Summary

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I. Introduction

This report covers the jurisdictions of **Australia**, **Brazil**, **Canada**, **European Union**, **India**, **Israel** and **Japan**. As European Union (EU) law regulates drones in the EU, the EU report also includes examples from **France**, **Germany**, and **Sweden**. The EU only allows member states to regulate matters not directly regulated or mandated in the regulation, including age requirements in the open and specified categories and fees charged.

While the United States uses categories such as commercial to define training requirements, not all jurisdictions surveyed distinguish operational and training requirements based on whether the use is commercial. For example, the EU classifies drone categories into “Open,” “Specific,” and “Certified.”

II. Training & Certification Process

A. Training Requirements

1. Qualifying Age Requirements

Eligibility for drone operations is typically dependent on the type of drone (size, weight) and manner of operation. In the case of smaller drones used for recreational purposes, minors are typically allowed to operate them. For example, **Sweden** allows the operation of drones in the open category from 15 whereas most other **EU** member states allow this operation from age 16 which is the same as the age requirement in **Japan**. Similarly, **Canada** allows certain use by age 14 but requires a person seeking a license to perform advanced operations to be at least 16 years old.

In order to be eligible to fly a drone commercially however, most jurisdictions require that the operator is at least 18 years old before pursuing relevant training and meeting certification requirements. For example, **Brazil** specifically requires that any use that is not for recreational use be done by persons 18 years or older. Also **India** and **Israel** require that the license holder be at least 18 years old, with **India** capping the age eligibility at 65 years old. Only **Australia** does not specify any upper or lower age limits as a precondition for pilot training.

2. Theoretical and Practical Training

To obtain a license to operate drones that are used commercially, theoretical and practical training are typically required in all jurisdictions surveyed. In **Australia**, the training includes both knowledge and practical skills, which are both tested by a certified training provider. Training with regards to aeronautical knowledge must include at least 15 hours with a training instructor. In **Israel**, tested topics include specifically, among others, “aviation law,” “air traffic provisions,”

“general technical knowledge,” “principles of meteorology,” “practical navigation principles,” and “safety procedures.” In addition, an applicant must have a minimum of 250 hours of relevant operational experience. The EU does not require practical training for subcategory A1 of its open category, and all other training requirements differ depending on the class identification and the operation category of the drone.

B. Testing

Similar to the training requirements mentioned above, most surveyed jurisdictions require both a knowledge test and practical skills test for licenses to operate drones commercially. The operation of most smaller drones, however, which in **the EU** are designated as falling in the open category, requires an online written knowledge test on the subjects of air safety, airspace restrictions, aviation regulation, human performance limitations, operational procedures, drone general knowledge, privacy and data protection, insurance, and security.

C. Fees

Fees and the fee structure for training and licensing of drone operations vary among the surveyed jurisdictions. Most jurisdictions require fees for taking the test, as well as for receiving the certification. For example, **Brazil** leverages separate fees for registration, testing, and the issuance of an airworthiness certificate. In **Israel**, different fees apply depending on drones’ weight for applications for certificate of fitness, with additional fees for experimental or special use of drones, as well as for operating drones. Member states of the **European Union**, such as **Sweden**, also leverage different fees depending on what category the license falls under, making licenses in the open category less expensive than in the specific category. In addition, **France** allows unemployed jobseekers to take the knowledge exam for free.

D. Certification/License

Most jurisdictions surveyed require that a person who completes the training, testing, and pays the relevant fees, also registers in a national registry before being permitted to operate drones.

III. Renewal Requirements

Most jurisdictions require renewal of licenses for operating drones both recreationally and commercially. For example, **Brazil** requires annual revalidations which include knowledge and practical tests. In **Israel**, licenses expire after 24 months but can be extended if they meet minimum time operational requirements. In **Japan**, the license must be renewed after three years and requires completion of additional theoretical training. In **the EU**, renewal requirements depend on the category the drone operation falls into, with the open category requiring renewal every five years, whereas the specific category does not expire but the certification, which states that the holder continues to comply with relevant rules, is only valid for two years. For the open category, a refresher training instead of retesting may be chosen if the renewal takes place within the validity period. Similarly, **India** requires refresher training for renewal at least every 10 years.

In **Canada**, pilot licenses do not expire but license holders must maintain compliance with the recency requirements, including knowledge exams and flight hours.

It appears most jurisdictions surveyed do not have an upper age limit for renewal of licenses. Of the surveyed jurisdictions, only **India** has an explicit upper age requirement, capping renewals at 65 years of age.

It appears **Australia** has no renewal requirements.

Australia

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SUMMARY People who wish to fly certain types of drones in Australia for commercial purposes are required to obtain a remote pilot license (RePL) issued by the Civil Aviation Safety Authority (CASA). Those who trade as drone service providers for hire or reward must hold a remotely piloted aircraft operator's certificate (ReOC) and can then employ RePL holders to operate drones for the business. To obtain a RePL, a person must take a training course provided by a certified RePL training provider and pass the required theory and practical skills tests for the type and category of RPA he or she intends to fly. The person must also log at least five hours of flying time with the relevant type of drone. Once the requirements have been met, the RePL training provider applies to CASA for a RePL on the person's behalf. The units, standards, and testing requirements for the training courses are set out in a legislative instrument. Different units apply for different categories of drone, and a RePL will be issued with respect to only the category that a person is qualified to fly. There are no age restrictions or renewal requirements for RePL.

I. Introduction

In Australia, remotely piloted aircraft (RPA) and those who operate or pilot them for commercial purposes are subject to various rules under part 101 of the Civil Aviation Safety Regulations 1988 (Cth) (CASR).¹ A Manual of Standards² (MOS) and other legislative instruments also apply, and the Civil Aviation Safety Authority (CASA) has produced guidance material on the rules.³

Holding a remotely piloted aircraft operator's certificate (ReOC) allows a person or business to trade as a drone service provider for hire or reward. A holder of an ReOC is permitted to conduct a range of RPA operations, to employ remote pilots to operate RPA, and is subject to certain obligations. The ReOC does not allow the holder to operate an RPA – in order to operate an RPA, a person must hold a remote pilot license (RePL).⁴

RPA are categorized under the CASR according to their size and type.⁵ An ReOC is not required if a person or business is eligible to operate a “micro RPA” (up to 250 grams) or RPA that fall

¹ Civil Aviation Safety Regulations 1988 (Cth) (CASR) (vol 3) pt 101, <https://perma.cc/H3RH-UCTT>.

² Part 101 (Unmanned Aircraft and Rockets) Manual of Standards 2019 (as amended) (MOS), <https://perma.cc/23VT-T4KN>.

³ See *Part 101 of CASR Unmanned Aircraft and Rockets*, Civil Aviation Safety Authority (CASA), <https://perma.cc/C8UU-XYWP>.

⁴ *Remotely Piloted Aircraft Operator's Certificate*, CASA, <https://perma.cc/KRU6-FXDA>. See CASR div 101.F.4.

⁵ *Types of Drones*, CASA, <https://perma.cc/9QT2-P8P9>; CASR vol. 5 pt 1.

within an excluded category under the regulations, or if the person or business is flying an RPA for recreation. ReOC holders can employ remote pilots to operate RPA that weigh less than 7 kilograms (kg), less than 25 kg, less than 150 kg, or more than 150 kg (with additional certification required for the RPA). The ReOC lists the conditions of operation, including the type of RPA, weight restrictions, and manufacturer and model (for RPA over 25 kg).⁶ ReOC are valid for 12 months and renewals are valid for up to three years.⁷

Any drone flown for business or as part of a person's job must be registered with CASA. A person must be aged 16 years or over to register a drone.⁸

People are able to operate drones of not more than 2 kg ("very small RPA") for commercial or professional activities for work, training, or community services, or to use a drone of not more than 25 kg ("small RPA") over their own land for certain purposes, without holding a RePL.⁹ Certain safety and registration rules apply.¹⁰

However, a person who flies micro or very small RPA for business or as part of his or her job, or who flies small RPA over his or her own land, must obtain a RPA operator accreditation (unless he or she holds a RePL). Such accreditation is free, can be completed online, and is valid for three years. A person must be aged 16 years or older to become an accredited operator.¹¹

This report provides information on the training required for a person to obtain a RePL in order to operate different categories of RPA.

II. Training and Certification Process for Remote Pilot Licenses

To obtain a RePL, a person must get an aviation reference number (ARN), take a training course provided by a certified RePL training provider, pass the required theory part of the RePL course, and pass the practical skills part of the course for the type and category of RPA he or she intends to fly. Once the tests are passed, the RePL training provider applies to CASA for a RePL on the person's behalf.¹² There are no age requirements for obtaining a RePL.¹³

RePL can also be obtained based on recognition of a person's aviation experience where he or she has either passed the theory part of any "Part 61 flight crew licence" or has an equivalent military qualification or air traffic control license. International qualifications "may be recognised

⁶ Remotely Piloted Aircraft Operator's Certificate, *supra* note 4.

⁷ *Id.*

⁸ *Register Your Drone*, CASA, <https://perma.cc/MNT8-2DWF>.

⁹ CASA, *CASR Part 101: Micro and Excluded Remotely Piloted Aircraft Operations – Plain English Guide 5 & 8* (Sept. 2020), <https://perma.cc/8F42-FACC>.

¹⁰ See *Drone Rules*, CASA, <https://perma.cc/8JXE-5YGJ>.

¹¹ *Operator Accreditation*, CASA, <https://perma.cc/UV8K-4KNX>.

¹² *Remote Pilot Licence*, CASA, <https://perma.cc/NS9V-J937>; CASR 101.290 & 101.295

¹³ *Remote Pilot Licence*, *supra* note 12.

following a review and comparison of the foreign and Australian syllabi.”¹⁴ Where a person’s experience is recognized, he or she does not need to sit the theory component of the RePL training course. He or she can either take a course with a certified RePL training provider and pass the skills test plus log five hours of flying time with the RPA, or can book a test flight with CASA and pass the practical skills part of the RePL competency as well as logging five hours of flying time.¹⁵

An applicant for a RePL must have general English language competency. A RePL training course may specifically include English language proficiency training and assessment. Otherwise, “applicants must obtain their general English training proficiency by virtue of holding, or having held, a flight crew license issued by CASA.”¹⁶

The CASR provisions applicable to RPA are contained in subpart 101.F, with RePL applications, eligibility, conditions, and variations, suspensions, and cancellations covered in division 101.F.3. Provisions related to ReOC are contained in division 101.F.4. In addition, part 11 of the CASR contains provisions relating to applications for RePL.¹⁷

The MOS for part 101 “defines a RePL Training Course and includes the required RePL instructor qualifications and requirements for operator documented practices and procedures.”¹⁸

A. Training Requirements

Division 2.2 of the MOS contains the standards and requirements for the theory component of the RePL training course (“aeronautical knowledge component”) and “the component of a RePL training course for the manual or automated operation of a category of RPA that an applicant for a RePL proposes to operate” (“practical competency component”).¹⁹

The schedules of the MOS set out detailed units and standards for each component of the RePL training courses with respect to different categories of RPA:

- Schedule 2 lists the aeronautical knowledge standards for a RePL training course, including common standards applicable to all categories and “units of knowledge” for the different categories: airplane, helicopter (multirotor class), helicopter (single rotor class), powered-lift, and RPA with a liquid fuel system.
- Schedule 3 lists the practical competency standards, again including common standards and “units of practical competency” for each of the above-listed categories.

¹⁴ Id.

¹⁵ Id.

¹⁶ MOS note to 2.03(1).

¹⁷ CASR regs 11.033 & 11.034 etc.

¹⁸ CASA, *Annex A to AC 101-01: Remote Pilot Licence (RePL) Training Course – CASA Guidance A6*, <https://perma.cc/HK6Y-DQF9>.

¹⁹ MOS s 2.02.

- Schedule 4 lists the aeronautical knowledge units, with common units broken into several units covering different areas of knowledge (e.g., basic aviation knowledge; airspace, charts, and aeronautical publications; basic meteorology for RPA operations; electrical and electronic systems for RPA systems; human performance; operations and procedures; operational rules and air law; and automated flight management systems). It also lists category-specific units for each of the RPA categories.
- Schedule 5 lists the practical competency units, including several common units and units for each RPA category. The category-specific units cover, for example, control on ground; launch and landing; hover (for certain categories); climb, cruise, and descent (for certain categories); normal operations; advanced maneuvers; and operation in abnormal situations and emergencies.
- Schedule 6 lists the flight test standards with respect to each of the categories of RPA, including the flight test requirements, knowledge requirements, and practical flight standards. Applicants must demonstrate their competency by performing the different maneuvers listed for the relevant category within the accuracy/tolerances specified for those maneuvers. They must also demonstrate their knowledge of certain matters with respect to the operation of the RPA in the relevant category.

The MOS requires that “[a] RePL training course for a category of RPA must include training and assessment in the units of aeronautical knowledge (including common units) that are for the category in accordance with the standards and requirements in Schedule 2.”²⁰ The training course must also require the applicant to complete specified units of knowledge contained in schedule 4.²¹

Similarly, “[a] RePL training course for a category of RPA must include training and assessment in the units of practical competency that are for the category in accordance with the standards and requirements in Schedule 3.”²² The course must require the applicant to complete specified units of practical competency contained in schedule 5.²³ In addition, the practical competency component of a course must be for either the automated operation mode (for an applicant for a RePL for automated mode only), or for both automated operation mode and the manual mode (for an applicant for a RePL for the manual operation mode).²⁴ Furthermore, “the practical competency component of a RePL training course for a particular medium or large RPA for which the applicant seeks the RePL must be conducted and assessed with respect to the particular medium or large RPA only.”²⁵

²⁰ Id. s 2.05(1).

²¹ Id. s 2.05(4).

²² Id. s 2.06(3).

²³ Id. s 2.06(6).

²⁴ Id. s 2.06(1).

²⁵ Id. s 2.06(2).

In relation to the aeronautical knowledge component of a RePL training course, a training organization must provide students with at least 15 hours of contact time with a RePL training instructor. Where the training course covers more than one category of RPA, four additional hours of contact time must be provided for each additional category.²⁶

The CASR require that, in order to be eligible for a RePL, an applicant must have at least five hours of experience in operating an RPA under standard operating conditions.²⁷ According to the MOS, to deliver a RePL training course for any type of RPA, a training organization must ensure that each student is provided with a means of achieving or simulating the flight conditions and variables mentioned in each item of the practical competencies for the relevant category in schedule 5 of the MOS. The amount of flying required for a category, size, or complexity of RPA must be specified in the training organizations documented practices and procedures.²⁸

B. Testing

1. Theory Component

Division 2.3 of the MOS applies to the testing of the RePL training course theory component. It states that

- (1) To pass the theory component of a RePL training course (the *aeronautical knowledge component*), an applicant must pass an examination (the *aeronautical knowledge examination*, or the *examination*).
- (2) The examination must be a closed-book examination except for any documents concerning the RPAS:
 - (a) authored by CASA and published on its website; or
 - (b) authored by AA [Airservices Australia] and published on its website.

Note The examination is one in which the candidate must not use any RePL training course document or material, separate from the examination text, to assist in answering examination questions. However, official, online CASA and AA documents may be used.²⁹

The pass mark for the examination is 85 percent, with the examination required to be assessed by a RePL training instructor or the chief remote pilot for the RePL training organization. If an applicant fails the examination three times, he or she can only re-sit it a fourth time if he or she has repeated the aeronautical knowledge component of the RePL training course and at least 14 days have elapsed since the third attempt.³⁰

²⁶ Id. s 2.26.

²⁷ CASR reg 101.295(2)(c).

²⁸ MOS s 2.28.

²⁹ Id. s 2.08.

³⁰ Id. s 2.09.

The examination “must be a set of at least 80 multiple-choice questions covering all of the aeronautical knowledge units in Schedule 4 that are for the relevant category of RPA, including for automated flight management systems and liquid-fuel systems if required (the *relevant units*), for which the applicant is applying for a RePL (the *relevant RePL*).”³¹ The MOS sets the minimum number of questions to be based on items in the relevant units.³² The duration of the examination must be the sum of 15 minutes plus one minute for each of the questions.³³ Each training organization must have at least four unique sets of questions that have been approved by CASA, with the examinations to be randomly rotated for each training course.³⁴

2. *Practical Component*

Division 2.4 of the MOS contains provisions on the assessment of the practical competencies for a RePL training course. It states as follows:

- (2) Subject to sections 2. and 2.19AB, to complete the RePL training course component for the operation of a category of RPA (the practical competencies), the applicant must be assessed as competent in each of the units in Schedule 5 (the relevant practical competency units) that is:
 - (a) for the relevant RPA category (including with or without a liquid-fuel system, as the case requires); and
 - (b) conducted:
 - (i) for the automated operation mode; or
 - (ii) for both the automated operation mode and the manual mode;as the case requires for the RePL that the applicant is applying for (the relevant RePL).
- (3) To be assessed as competent, the applicant must:
 - (a) demonstrate to a RePL training instructor all of the behaviours mentioned in each item of the relevant practical competency unit; and
 - (b) satisfy the RePL training instructor that each of the behaviours referred to in paragraph (a) has been demonstrated within the relevant tolerances, and across the range of variables, (if any), mentioned for the item; and
 - (c) having satisfied the requirements of paragraphs (a) and (b), pass a RePL training course flight test in the relevant RPA, conducted in accordance with the RePL training organisation’s documented practices and procedures by an examiner.
- (4) Subject to subsection (5), for subsection (3), a behaviour must be demonstrated:
 - (a) for the relevant RPA category and the relevant RePL; and
 - (b) across as many of the range of variables as the operating conditions reasonably permit.
- (5) If a variable was not selected for demonstration because operating conditions made it impossible in practice to demonstrate the variable, the RePL training instructor must:
 - (a) require the applicant to provide:
 - (i) a satisfactory computerised simulation of the flight characteristics of the RPA under the variable, using a computerised simulation system that meets the

³¹ Id. s 2.10(1).

³² Id. s 2.10(2).

³³ Id. s 2.11.

³⁴ Id. s 2.12.

- standards and requirements set out in the operator's documented practices and procedures; or
- (ii) a detailed oral explanation of the variable, accompanied by indicative manual use of the RPA's functional elements, or a description of such use; and
- (b) certify in writing to the RePL training organisation, in accordance with the organisation's documented practices and procedures, that:
 - (i) operating conditions made it impossible in practice to demonstrate the variable; and
 - (ii) for reasons stated in the certification, it would be unreasonable to require that the variable be demonstrated on another occasion.
- (6) For paragraph (3) (b), a behaviour demonstrated outside a relevant tolerance may be considered to have been demonstrated within the tolerance if the RePL training instructor:
 - (a) is satisfied that:
 - (i) the only reason for failure to demonstrate the behaviour is the unavoidable impact of the relevant operating conditions; and
 - (ii) in the circumstances, the behaviour that was demonstrated was not so far outside the tolerance as to indicate a lack of competence in the relevant operating conditions; and
 - (b) certifies to the RePL training organisation, in accordance with the organisation's documented practices and procedures, that paragraph (a) was the case.
- (7) The training course flight test for the relevant RPA mentioned in paragraph (3) (c) must be in accordance with the flight test standards in Schedule 6.³⁵

Division 2.8 of the MOS contains additional provisions related to flight tests. It states that a student enrolled with an RPA training organization "may attempt the flight test for a RePL at any time during the RePL training course," provided the attempt meets the requirements.³⁶

In addition, as noted above, the MOS states that a RePL training course may include, as part of the practical competency component, "training and assessment in general English language proficiency."³⁷

C. Fees

Fees for RePL and other services related to unmanned aircraft are set under part 17 of schedule 1 of the Civil Aviation (Fees) Regulations 1995 (Cth). An hourly rate is charged by CASA for the granting of a RePL, including processing and consideration of an application.³⁸ The regulations provide that the hourly rate for this service is AU\$160 (approximately US\$111).³⁹

³⁵ Id. s 2.18.

³⁶ Id. 2.31(1).

³⁷ Id. s 2.03(1).

³⁸ Civil Aviation (Fees) Regulations 1995 (Cth) sch 1 item 17.5, <https://perma.cc/EU67-CD5G>.

³⁹ Id. reg 4(2)(c). See also *Service Standards and Fees*, CASA, <https://perma.cc/R5CY-YA9F>.

CASA provides a directory of ReOC holders, including those certified to conduct RePL training.⁴⁰ CASA has provided detailed guidance on training provider certification processes and course requirements as part of an advisory circular on part 101 of the CASR.⁴¹ It appears that training providers can set their own fees for the RePL training courses, which include the assessment process and may include the RePL application cost.⁴²

D. Certification/License

Each student who successfully completes a RePL training course must be given a certificate of course completion by the training organization.⁴³ As noted above, the training organization applies to CASA for a RePL on behalf of the student. The CASR provide for conditions to be placed on RePL, including conditions that allow the person to operate RPA of only specified kinds, limit the areas where he or she may operate RPA, or allow him or her to operate RPA only in visual meteorological conditions.⁴⁴ In addition,

- (3) It is a condition of a remote pilot licence that the licence holder must not operate an RPA above 400 ft AGL in controlled airspace, or within 3 nautical miles of the movement area of a controlled aerodrome, unless he or she holds at least one of the following qualifications:
 - (a) an aeronautical radio operator certificate;
 - (b) a flight crew licence;
 - (c) an ATC licence;
 - (d) a military qualification equivalent to a licence mentioned in paragraph (b) or (c);
 - (e) a flight service licence.
- (4) It is a condition of a remote pilot licence that an RPA must be operated within the visual line of sight of the licence holder unless he or she:
 - (a) has passed:
 - (i) an aeronautical knowledge examination (within the meaning of Part 61) for the grant of an instrument rating under Part 61; or
 - (ii) an aviation licence theory examination before 1 September 2014 that is taken to be an equivalent requirement for the grant of an instrument rating under regulation 202.274; or
 - (iii) an approved examination; and
 - (b) either:
 - (i) holds both a certificate as an RPA operator under Division 101.F.4 and an approval under regulation 101.029 to operate the RPA beyond the person's visual line of sight; or
 - (ii) is a member of an RPA operator's personnel and the RPA operator holds both a certificate as an RPA operator under Division 101.F.4 and an approval under

⁴⁰ Remotely Piloted Aircraft Operator's Certificate (ReOC) Holders, CASA, <https://perma.cc/PN6A-KUR2>.

⁴¹ Annex A to AC 101-01: Remote Pilot Licence (RePL) Training Course – CASA Guidance, *supra* note 18.

⁴² See, e.g., *Remote Pilot Licence Multirotor 25kgs*, Avisssist, <https://perma.cc/64HP-WH2X>; *Remote Pilot Training Courses*, Aviation Australia, <https://perma.cc/6XUH-PNS9>; *CASA Drone Certification – Initial RePL Training*, UAV Training Australia, <https://perma.cc/MV93-BKJ3>.

⁴³ MOS s 2.29.

⁴⁴ CASR reg 101.300(2).

regulation 101.029 for the operator's personnel to operate RPA beyond their visual line of sight.

- (5) It is a condition of a remote pilot licence that the licence holder must not operate more than 1 RPA at a time unless:
 - (a) he or she holds an approval under regulation 101.029 to operate more than 1 RPA at a time; and
 - (b) the conditions (if any) imposed on the approval are complied with.⁴⁵

III. Renewal Requirements

There are no renewal requirements for RePL.

⁴⁵ Id. reg 101.300(3)–(5).

Brazil

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SUMMARY In Brazil, the operation of model aircrafts does not require any specific licensing or qualification, but must be confined to appropriate spaces; they cannot fly above 400 feet and can weigh up to 250 gr.

Remotely-piloted aircrafts, commonly known as drones, are divided into three classes according to their weight. A person must be 18 years old, have a license, medical certificate, and carry third-party insurance to operate such devices. The equipment must be registered with the telecommunications agency and, depending on its class, flights must be registered with the entity responsible for the Brazilian airspace. The pilot must comply with the regulations of the civil aviation agency.

I. Introduction

A. Definition

According to the Center for Investigation and Prevention of Aeronautical Accidents of the Air Force Command in Brazil, “drone” is considered to be a generic term without technical support or legal definition to characterize any and all unmanned flying objects, be it of any purpose (professional, recreational, military, commercial, etc.), origin, or characteristic.¹

Unmanned Aerial Vehicle (*Veículo Aéreo Não Tripulado*, VANT) is the official terminology provided by Brazilian air transport regulatory bodies to define these objects.²

Air Information Circular AIC N 21/10 of September 23, 2010, issued by the Department of Airspace Control (*Departamento de Controle do Espaço Aéreo*, DECEA) for the purpose of presenting the necessary information for the use of these devices in Brazilian airspace,³ defines a VANT as an aerial vehicle designed to operate without a pilot on board, which has a payload on board and that is not used for purely recreational purposes.⁴ In this definition, all three-axis controllable airplanes, helicopters, and airships are included, excluding, therefore, traditional balloons and model aircrafts (*aeromodelos*).⁵

¹ DECEA *Esclarece Normas para Voos de Drones no Brasil*, Centro de Investigação e Prevenção de Acidentes Aeronáuticos, Comando da Aeronáutica, <https://perma.cc/NP62-KCJS>.

² Id.

³ Circular de Informações Aeronáuticas AIC No. 21/10, de 23 de Setembro de 2010, Section 1.1, <https://perma.cc/XU6D-8Q4G>.

⁴ Id. Section 2.2.22.

⁵ Id. Model aircrafts are regulated by Portaria DAC No. 207/STE, de 7 de Abril de 1999, <https://perma.cc/2YKK-KPGY>.

There are two types of VANTs: remotely-piloted aircrafts, RPAs (*aeronaves remotamente pilotadas*), in which the pilot is not on board⁶ and autonomous aircrafts (*aeronaves autônomas*) that, once programmed, do not allow external intervention during the flight.⁷

The best-known VANT is the RPA in which the pilot controls the aircraft remotely from any interface (computer, simulator, digital device, remote control, etc.);⁸ unlike an autonomous aircraft that does not allow external intervention.⁹ In Brazil, autonomous aircrafts are prohibited from being used in the country.¹⁰

B. Regulatory Bodies

1. National Telecommunications Agency

The National Telecommunications Agency (*Agência Nacional de Telecomunicações*, ANATEL) was created by Law No. 9,472, of July 16, 1997,¹¹ with the function of regulatory body for telecommunications and is subordinated to the Ministry of Communications.¹²

2. Department of Airspace Control

The DECEA is the organization responsible for controlling Brazilian airspace, providing air navigation services that enable flights, and the ordering of air traffic flows in the country.¹³

On May 22, 2020, DECEA enacted Ordinance No. 112/DGCEA, which approved the reenactment of ICA 100-400, Instruction on Unmanned Aircrafts and the Access to Brazilian Airspace (the Instruction).¹⁴ The purpose of the Instruction is to regulate the procedures and responsibilities necessary for safe access to Brazilian airspace by unmanned aircrafts.¹⁵

The Instruction determines that it is the responsibility of the DECEA, as the central body of the Brazilian Airspace Control System (*Sistema de Controle do Espaço Aéreo Brasileiro*, SISCEAB), to

⁶ Circular de Informações Aeronáuticas AIC No. 21/10, Section 2.2.3.

⁷ Id. Section 2.2.1.

⁸ DECEA *Esclarece Normas para Voos de Drones no Brasil*, Centro de Investigação e Prevenção de Acidentes Aeronáuticos, *supra* note 1.

⁹ Id.

¹⁰ Id.

¹¹ Lei No. 9.472 de 16 de Julho de 1997, art. 8, <https://perma.cc/Z9RZ-AJJB>.

¹² Decreto No. 2.338 de 7 de Outubro de 1997, ANEXO I, art. 1, <https://perma.cc/24SN-C4PS>.

¹³ O DECEA, Ministério da Defesa, Departamento de Controle do Espaço Aéreo, <https://perma.cc/VQE4-EXDT>.

¹⁴ Portaria DECEA No 112/DGCEA, de 22 de Maio de 2020, aprova a reedição da ICA 100-40, Instrução sobre “Aeronaves não tripuladas e o Acesso ao Espaço Aéreo Brasileiro,” <https://perma.cc/ZQU6-QG7J>.

¹⁵ Id. Section 1.1.

legislate on the procedures for access to the airspace, and that the other regulatory bodies are responsible for dealing with subjects of their respective areas of activity.¹⁶

According to the Instruction, an aircraft is any device that can sustain itself in the atmosphere from reactions of the air other than the reactions of the air against the earth's surface.¹⁷ An aircraft that does not have a crew on board is an aircraft and, therefore, to fly in the airspace which falls under the responsibility of Brazil, it must follow the rules established by the competent authorities of national aviation.¹⁸ The remote pilot is ultimately responsible for the faithful observance and fulfillment of all the parameters provided for in Instruction ICA 100-40.¹⁹

3. *National Civil Aviation Agency*

The National Civil Aviation Agency (*Agência Nacional de Aviação Civil, ANAC*) was created by Law No. 11,182 of September 27, 2005²⁰ and its purpose is to regulate and inspect civil aviation and aeronautical and airport infrastructure activities.²¹

C. **Classes of RPAs**

RPAs are divided into three classes, according to the maximum take-off weight, in which the weights of the battery or fuel of the equipment and any cargo eventually transported must be considered.²² The classification is only applicable to RPAs and not to model aircrafts.²³ The classes are as follows:

- Class 1 – Maximum take-off weight greater than 150 kg (330.7 lb)
- Class 2 – Maximum take-off weight greater than 25 kg and up to 150 kg (55.1 lb – 330.7 lb)
- Class 3 – Maximum take-off weight up to 25 kg (55.1 lb)
 - Model aircraft or RPA with a maximum take-off weight of up to 250g (0.55lb)
 - RPA with maximum take-off weight greater than 250g and up to 25 kg (0.55 lb – 55.1 lb).²⁴

¹⁶ Id. Section 1.2.

¹⁷ Id. Section 4.1.

¹⁸ Id. Section 4.7.

¹⁹ Id. Section 4.13.

²⁰ Lei No. 11,182, de 27 de Setembro de 2005, art. 1, <https://perma.cc/5E3D-LSLL>.

²¹ Decreto No. 5.731 de 20 de Março de 2006, ANEXO I, art. 1, <https://perma.cc/BWY4-JR5C>.

²² *Classes de Drones (RPA)*, Agência Nacional de Aviação Civil (ANAC), <https://perma.cc/G67K-FXZD>.

²³ Id.

²⁴ Id.

D. Documentation

1. Class 1

To operate a Class 1 unmanned aircraft, in addition to following ANATEL's and DECEA's regulations, the following are necessary²⁵:

- To be at least 18 years of age to fly or assist the operation as an observer;
- To possess insurance with third-party damage coverage;
- To conduct an operational risk assessment (IS-ANAC No. E94-003²⁶);
- To operate only in areas far from third parties (minimum 100 horizontal feet). This restriction is waived if there is consent from people close to the operation or there is a mechanical barrier capable of isolating and protecting people not involved and not consenting to the operation (flight);
- To only operate a single RPA system²⁷ at a time;
- To allow a change of the remote pilot in command during operation;
- To start operations only if there is sufficient autonomy of the aircraft to perform the flight and to land safely at the intended location, taking into account the known meteorological conditions;
- To possess and carry a license (*licença e habilitação*) issued by ANAC, as well as an aeronautical medical certificate²⁸ (*Certificado Médico Aeronáutico*, CMA) of 1st, 2nd, or 5th class granted by ANAC, or 3rd class issued by the Air Force Command;
- To obtain registration with ANAC and carry a standard or restricted Certificate of Airworthiness (*Certificado de Aeronavegabilidade*, CA). In this case, the owner must follow all procedures established in RBAC n° 21.²⁹ In specific situations or in the case of aircraft for experimental purposes, the standard or restricted CA may be replaced by an experimental

²⁵ *Drones Classe 1 (RPA com Peso Máximo de Decolagem Maior que 150 Kg)*, ANAC, <https://perma.cc/NF3Y-6BPB>.

²⁶ ANAC Instrução Suplementar IS No. E94-003, Revisão A, <https://perma.cc/45UD-Q2VV>.

²⁷ *DECEA Esclarece Normas para Voos de Drones no Brasil*, Centro de Investigação e Prevenção de Acidentes Aeronáuticos, *supra* note 1. An RPA system is defined as all the resources of the system that makes it fly: the remote piloting station, the link or command link that enables the control of the aircraft, its support equipment, etc.

²⁸ ANAC Regulamento Brasileiro de Aviação Civil (RBAC) No. 67, as amended by Emenda 05, Section 67.3(a), <https://perma.cc/8NGE-P8PW>, defines an aeronautical medical certificate as a certificate issued by ANAC, after examination health tests are carried out on candidates, attesting to their psychophysical aptitudes, in accordance with RBAC No. 67, to perform duties related to aircrafts. The CMA is equivalent to the physical capacity certificate (*Certificado de Capacidade Física*, CCF) for the purpose of complying with the rules contained in articles 159 to 164 and article 302 of the Brazilian Aeronautical Code, Código Brasileiro de Aeronáutica, Lei No. 7.565, de 19 de Dezembro de 1986, <https://perma.cc/5ESL-MH7J>, and of article 72 of the Aeronaut Law, Lei do Aeronauta, Lei No. 13.475, de 28 de Agosto de 2017, <https://perma.cc/925Z-TKA2>.

²⁹ ANAC RBAC No. 21, as amended by Emenda 08, Section 21.1(a)(1)(iii), <https://perma.cc/Z3CL-QJZJ>, establishes procedural requirements for the issuance and modifications of, among other things, certificates of airworthiness.

flight authorization certificate (*Certificado de Autorização de Voo Experimental*, CAVE³⁰) or a Special Flight Authorization (*Autorização Especial de Voo*, AEV³¹);

- To perform an annual maintenance inspection (*Inspecção Anual de Manutenção*, IAM³²) on the RPA system every 12 months;
- For the owner or operator to submit to ANAC a declaration of annual maintenance inspection (*Declaração de Inspecção Anual de Manutenção*, DIAM) for the referred RPA system, attesting to its airworthiness condition. Maintenance, preventive maintenance, repairs or alterations must be performed as set out in the applicable requirements of Subpart G of ANAC RBAC-E No. 94/2017³³ and other applicable regulations, including ANAC RBAC No. 43;³⁴
- To not operate unmanned aircrafts under the influence of psychoactive substances. All operators are subject to the rules regarding the use of alcohol and drugs contained in Section 91.17 of RBAC 91.³⁵

2. Class 2

To operate a Class 2 unmanned aircraft, in addition to following ANATEL and DECEA regulations, it is necessary³⁶

- To be at least 18 years of age to fly or assist the operation as an observer;
- To possess insurance with third-party damage coverage;
- To conduct an operational risk assessment (IS-ANAC No. E94-003³⁷);

³⁰ Id. Sections 21.191, 21.193, 21.195, 21.211, 21.223.

³¹ Id. Sections 21.197, 21.199.

³² ANAC Instrução Suplementar IS N° 91.403-001, Revisão C, Section 4.4, <https://perma.cc/LS2E-QXGR>, defines annual inspection as the inspection required in Section 91.409(a)(1) of ANAC RBAC No. 91 with the scope detailed in Appendix D of RBAC No. 43. Section 91.409(a)(1) of ANAC RBAC No. 91, as amended by Emenda 03, <https://perma.cc/TWR5-LBUA>, determines that an aircraft is only allowed to be operated if, within the 12 months preceding the operation, this aircraft has been subjected to an annual inspection in accordance with RBAC No. 43 and approved for return to service by a person authorized by Section 43.7 of RBAC No. 43. Section 43.7 (a) of ANAC RBAC No. 43, as amended by Emenda 05, <https://perma.cc/S385-PCST>, states that only ANAC or a person in compliance with the provisions of Section 43.7 and 43.17 may approve the return to service of an article that has been subjected to maintenance, preventive maintenance, reconstruction and alteration.

³³ ANAC Regulamento Brasileiro de Aviação Civil Especial (RBAC-E) No. 94, as amended by Emenda No. 02, Preamble, <https://perma.cc/ZU4E-QAR8>, establishes the conditions for the operation of unmanned aircrafts in Brazil considering the current stage of development of this technology.

³⁴ Section 43.1 (a)(1) of ANAC RBAC No. 43, establishes rules for maintenance, preventive maintenance, rebuilding and alteration, including major repairs and major alterations of any aircraft that have a Brazilian certificate of airworthiness.

³⁵ *Drones Classe 1 (RPA com Peso Máximo de Decolagem Maior que 150 Kg)*, ANAC, supra note 25.

³⁶ *Drones Classe 2 (RPA com Peso Máximo de Decolagem Maior que 25 Kg e até 150 Kg)*, ANAC, <https://perma.cc/2HVB-9895>.

³⁷ ANAC Instrução Suplementar IS No. E94-003, Revisão A, <https://perma.cc/45UD-Q2VV>.

- To operate only in areas far from third parties (minimum 100 horizontal feet). This restriction is waived if there is consent from people close to the operation or there is a mechanical barrier capable of isolating and protecting people not involved and not consenting to the operation;
- To only operate a single RPA system at a time;
- To be able to change the remote pilot in command during operation;
- To start remote operations only if there is sufficient autonomy of the aircraft to perform the flight and to land safely at the intended location, taking into account the known meteorological conditions;
- To obtain registration of the RPA with ANAC and carry an RPA special airworthiness certificate (*Certificado de Aeronavegabilidade Especial para RPA, CAER*³⁸). In this case, the owner must ask the manufacturer for a declaration that that specific RPA system is in accordance with the design authorized by ANAC. This declaration must be presented when the request for issuance of the certificate is made. The RPA system must be previously inspected by ANAC for registration and issuance of the CAER. In specific situations or in the case of aircrafts with experimental purposes, the CAER can be replaced by an experimental flight authorization certificate (*Certificado de Autorização de Voo Experimental, CAVE*) or a special flight authorization (*Autorização Especial de Voo, AEV*);
- To carry out all the specific procedures recommended by the manufacturer in the maintenance manual and record them in an appropriate notebook. Maintenance, preventive maintenance, repairs or alterations and approvals for return to service must be carried out by the manufacturer or a maintenance organization accredited by the manufacturer or by a person qualified and properly trained by the manufacturer or an institution accredited by the manufacturer;
- To possess and carry a license (*licença e habilitação*) issued by ANAC, as well as an aeronautical medical certificate³⁹ (*Certificado Médico Aeronáutico, CMA*) of 1st, 2nd or 5th class granted by ANAC or 3rd class issued by the Air Force Command;
- To not operate unmanned aircrafts under the influence of psychoactive substances. All operators are subject to the rules regarding the use of alcohol and drugs contained in Section 91.17 of RBAC 91.⁴⁰

3. Class 3

a. Model Aircrafts or RPAs with a Maximum Take-Off Weight of Up to 250g

According to DECEA, Ordinance DAC No. 207/STE of April 7, 1999, establishes the rules for model aircrafts in the country.⁴¹ Article 1(a) states that the operation of model airplanes must be carried out in locations sufficiently distant from densely populated areas.⁴² The operation of

³⁸ ANAC RBAC-E No. 94, Section E94.501 states no unmanned aircraft may fly without a valid certificate of airworthiness, *supra* note 33.

³⁹ ANAC RBAC No. 67, Section 67.3(a).

⁴⁰ *Drones Classe 2 (RPA com Peso Máximo de Decolagem Maior que 25 Kg e até 150 Kg)*, *supra* note 36.

⁴¹ *DECEA Esclarece Normas para Voos de Drones no Brasil*, Centro de Investigação e Prevenção de Acidentes Aeronáuticos, *supra* note 1.

⁴² Portaria DAC No. 207/STE, de 7 de Abril de 1999, art 1 (a), <https://perma.cc/BK95-6BAA>.

motorized model aircrafts in the vicinity of noise-sensitive areas of urban facilities, such as hospitals, religious temples, schools, and nursing homes, should be avoided.⁴³

Unless authorized, no model aircraft should be operated more than 400 feet above the earth's surface. The operation of model aircrafts in the vicinity of aerodromes may only be carried out after authorization of the person responsible for the operation of the aerodrome.⁴⁴ It is prohibited to operate model aircrafts in the areas of approach and take-off of aerodromes.⁴⁵ Operations with radio-controlled equipment that are not for sport and leisure must be submitted to the Civil Aviation Department (*Departamento de Aviação Civil*, DAC).⁴⁶

If a model aircraft is operated outside these areas, it becomes necessary, in addition to the requirements mentioned, for the pilot to be registered, and to share flight information, with SARPAS, a system developed by the DECEA for the purpose of enabling the request for access to Brazilian airspace for the use of RPAs.⁴⁷ For regular RPA (non-recreational use) operation, authorizations from ANATEL, ANAC and DECEA are required.⁴⁸

Model Airplanes and RPAs up to 250g, although they belong to Class 3, are exempt from several requirements.⁴⁹ For these items of equipment, operations (flights) are allowed by ANAC under the full responsibility of its pilot.⁵⁰ To operate them, it is necessary to follow ANATEL's rules and DECEA's rules for the use of airspace, and observe the following:

- The equipment does not need to be registered with ANAC;
- The risk assessment of the operation is waived, but the conditions of the aircraft regarding flight safety must be verified. The person must be aware of all the information necessary for the planning of the flight before starting it and must be prepared to act in all phases of the flight during the operation;
- There is no restriction on the minimum age to operate model aircrafts;
- The minimum age to operate an RPA, however, is 18 years;
- Pilots do not need a document issued by ANAC and are considered duly licensed;
- Only one RPA system can be operated at a time;
- It is not mandatory to have insurance with coverage for damages to third parties;

⁴³ Id.

⁴⁴ Id. art. 1 (c).

⁴⁵ Id. art. 1 (d).

⁴⁶ Id. art. 1 (e).

⁴⁷ Ministério dos Transportes, Portos e Aviação Civil, *Guia para Fiscalização da Operação de Aeronaves Não Tripuladas, Versão 4*, at 10, <https://perma.cc/365A-G7YK>.

⁴⁸ Id.

⁴⁹ *Drones (Aeromodelos ou RPA com Peso Máximo de Decolagem de até 250 g)*, ANAC, <https://perma.cc/ZQV9-VVBV>.

⁵⁰ Id.

- It is allowed to change the remote pilot in command during the operation;
- It is not necessary to register flights;
- Remote operations can only be started if there is sufficient autonomy of the aircraft to perform the flight and to land safely at the intended location, taking into account the known meteorological conditions;
- A person is not allowed to operate unmanned aircrafts under the influence of psychoactive substances and all operators are subject to the rules regarding the use of alcohol and drugs contained in Section 91.17 of RBAC 91.⁵¹

b. RPAs with Maximum Take-Off Weight Greater than 250g and Up to 25 kg

According to the Guide for Inspection of Operation of Unmanned Aircrafts issued by the Ministry of Transport, Ports and Civil Aviation, for the regular operation of model aircrafts (recreational use) with maximum weight greater than 250 grams, it is necessary to register the aircraft with the Unmanned Aircraft System (*Sistema de Aeronaves Não Tripuladas*, SISANT), which is a system created by the National Civil Aviation Agency (Agência Nacional de Aviação Civil, ANAC) to allow operators and owners to register equipment; and to obtain the approval of the equipment by the National Telecommunications Agency (*Agência Nacional de Telecomunicações*, ANATEL).⁵²

Preferably, the model aircraft should be used within the areas considered adequate to the practice of model airplanes, although they are not restricted to the same.⁵³

To operate Class 3 unmanned aircrafts, in addition to following ANATEL's rules and DECEA's rules for the use of airspace, it is necessary

- To be at least 18 years of age to fly or assist the operation as an observer;
- To possess insurance with third-party damage coverage;
- To conduct an operational risk assessment (IS-ANAC No. E94-003);
- To operate only in areas far from third parties (minimum 30 meters horizontally). This restriction is waived if there is consent from people close to the operation or there is a mechanical barrier capable of isolating and protecting people not involved and not consenting to the operation;
- To only operate a single RPA system at a time;
- To be able to change the remote pilot in command during operation;
- To start operations only if there is sufficient autonomy of the aircraft to perform the flight and to land safely at the intended location, taking into account the known meteorological conditions;
- To register each piece of equipment in ANAC's Unmanned Aircraft System (SISANT);

⁵¹ Id.

⁵² Ministério dos Transportes, Portos e Aviação Civil, *Guia para Fiscalização da Operação de Aeronaves Não Tripuladas*, Versão 4, supra note 47, at 10.

⁵³ Id.

- To place the identification of the equipment (number obtained in the registration process) in a visible place on the aircraft and to ensure the identification is made with non-flammable material;
- To carry proof of registration with ANAC, insurance and risk assessment and the equipment's flight manual;
- To not operate unmanned aircrafts under the influence of psychoactive substances. All operators are subject to the rules regarding the use of alcohol and drugs contained in Section 91.17 of RBAC 91.⁵⁴

II. Training & Certification Process

Brazilian Civil Aviation Regulation ANAC-RBAC No. 61 establishes the rules and procedures related to the granting of licenses, qualifications (*habilitações*), and certificates for pilots; the minimum requirements and standards that must be met for the granting and revalidation of these documents; and the prerogatives and limitations related to each license, qualification, or certificate.⁵⁵

A. Training Requirements and Testing

Training requirements and testing are discussed below in Part D, subparts 4 & 6. As discussed above, the minimum age to pilot RPAs is 18 years.

B. Fees

ANAC Resolution No. 653 of December 20, 2021, establishes the rules for the control and collection of revenue from the collection of the Civil Aviation Inspection Fee (*Taxa de Fiscalização da Aviação Civil*, TFAC).⁵⁶ The fee for the granting, renewal or registration of licenses, qualifications or certification of civil aviation personnel is R\$150,00 (about US\$27.75);⁵⁷ the fee for the registration for the theoretical exam of a civil aviation professional is R\$200,00 (about US\$37.00);⁵⁸ and the fee for the issuance of an airworthiness certificate is R\$1.500,00 (about US\$277.52).⁵⁹

C. Certification

1. Recreational Use

The National Telecommunications Agency (*Agência Nacional de Telecomunicações*, ANATEL) regulates the use of the frequency emitted by the radio control that is used to control a VANT.

⁵⁴ Id.

⁵⁵ ANAC RBAC No. 61, Subpart A, Section 61.1(a), <https://perma.cc/7VVX-69GM>.

⁵⁶ ANAC Resolução No. 653, de 20 de Dezembro de 2021, art. 1, <https://perma.cc/W5P8-UHM6>.

⁵⁷ Id. ANEXO, Item (a).

⁵⁸ Id. ANEXO, Item (b).

⁵⁹ Id. ANEXO, Item (v).

For the operation of unmanned aircrafts, it is necessary to obtain the approval of the radiofrequency module and remote control with the regulatory agency.⁶⁰

Considering that current VANTs already have their frequencies factory-approved, it is not necessary to follow any procedure. However, if a VANT has not been approved by ANATEL, it is necessary to request its approval. ANATEL approvals are regulated by the resolutions below.⁶¹

a. Equipment

According to the Guide for Inspection of Operation of Unmanned Aircrafts (Guide), the following legislation regulates the equipment used for recreational purposes:

- Resolution ANATEL No. 715 of October 23, 2019, provides the regulation for conformity Assessment and Approval of Telecommunications Products;⁶²
- Resolution ANATEL No. 680 of June 27, 2017, provides the regulation on restricted radiation radio communication equipment;⁶³
- Resolution ANATEL No. 635 of May 9, 2014, provides the regulation on authorization for the temporary use of radio frequencies;⁶⁴
- DECEA Ordinance No. 113/DGCEA of May 25, 2020, which approves Air Force Command Manual No. 56-2/2020 for the purpose of regulating the procedures and responsibilities necessary for access to Brazilian air space by unmanned aircrafts, with use exclusively for recreation, the so-called model aircrafts and replaces DECEA Air Information Circular AIC N 17/18 of June 11, 2018.⁶⁵

b. Pilots

According to the Guide, no authorization is required from ANATEL for pilots; and no flight authorization from ANAC is required, when flying up to 400 feet above ground. Outside of this area, a remote pilot license (*licença e habilitação de piloto remoto*) is required. DECEA does not require a registration for flights within the appropriate areas. Outside of these areas, a registration with SARPAS is required.⁶⁶

⁶⁰ Ministério dos Transportes, Portos e Aviação Civil, *Guia para Fiscalização da Operação de Aeronaves Não Tripuladas*, Section 3.3, *supra* note 47, at 18.

⁶¹ *Id.* Equipamentos, *Normas Aplicáveis*, at 28.

⁶² Resolução ANATEL No. 715, de 23 de Outubro de 2019, <https://perma.cc/XDM8-Y378>, revoked Resolução ANATEL No. 242, de 30 de Novembro de 2000.

⁶³ Resolução ANATEL No. 680, de 27 de Junho de 2017, <https://perma.cc/BRX2-RQZD>, revoked Resolução ANATEL No. 506, de 1 de Julho de 2008.

⁶⁴ Resolução ANATEL No. 635, de 9 de Maio de 2014, <https://perma.cc/84JZ-KDZ3>.

⁶⁵ Circular de Informações Aeronáuticas AIC No. 17/18, de 11 de Junho de 2018, Section 1.1, <https://perma.cc/9Q8D-8BB4>.

⁶⁶ Ministério dos Transportes, Portos e Aviação Civil, *Guia para Fiscalização da Operação de Aeronaves Não Tripuladas, Versão 4, Piloto*, at 28, *supra* note 47.

2. *Non-Recreational Use*

a. Equipment

The Guide defines non-recreational use as any and all use other than sport and leisure.⁶⁷ For the non-recreational use of VANTs, it is necessary to receive the approval of the equipment by ANATEL and a stamp certifying its approval.⁶⁸ The aircraft must have identification of the SISANT code or nationality and registration marks.⁶⁹ No authorization for the use of the equipment is required by DECEA.⁷⁰

ANATEL Resolutions Nos. 715, 680, and 635 (listed above) are also applicable to equipment for non-recreational use.⁷¹

Pursuant to the Guide,⁷² equipment for non-recreational use is subject to the Brazilian Aeronautical Approval Regulation RBHA No. 91, which contains the general rules for operation of civil aircrafts;⁷³ the Brazilian Civil Aviation Regulation RBAC No. 21, which regulates the certification of aeronautical products;⁷⁴ and Brazilian Special Civil Aviation Regulation RBAC-E No. 94, which establishes general requirements for civil use of unmanned aircrafts.⁷⁵

b. Pilots

For pilots, the Guide states that no registration is required from ANATEL; an authorization from ANAC is required; and a registration with SARPAS is also required.⁷⁶

In addition, the following regulations are also applicable to pilots:⁷⁷ Brazilian Aeronautical Approval Regulation RBHA No. 91, which contains the general rules for operation of civil aircrafts;⁷⁸ Brazilian Special Civil Aviation Regulation RBAC-E No. 94, which establishes general requirements

⁶⁷ Id. *Uso Não Recreativo* at 30.

⁶⁸ Id. *Equipamento, Autorizações Necessárias*.

⁶⁹ Id.

⁷⁰ Id.

⁷¹ Id. *Equipamentos, Normas Aplicáveis* at 30.

⁷² Id.

⁷³ ANAC Regulamento Brasileiro de Homologação Aeronáutica (RBHA) No. 91, as amended by Resolução No. 478, de 7 de Junho de 2018, <https://perma.cc/XTR2-AMX3>.

⁷⁴ ANAC RBAC No. 21.

⁷⁵ ANAC RBAC-E No. 94.

⁷⁶ Ministério dos Transportes, Portos e Aviação Civil, *Guia para Fiscalização da Operação de Aeronaves Não Tripuladas, Versão 4, Piloto, Normas Aplicáveis*, supra note 47, at 30.

⁷⁷ Id.

⁷⁸ ANAC RBHA No. 91.

for civil use of unmanned aircrafts;⁷⁹ and the Brazilian Civil Aviation Regulation RBAC No. 67, which establishes the requirements for granting of aeronautical medical certificates, for registration and accreditation of doctors, accreditation of clinics, and for agreements with public entities.⁸⁰

D. Licenses

1. Definitions

RBAC No. 61 defines qualification as an authorization associated with a license or certificate, in which the qualifications and respective validities, special operating conditions, and the respective attributions and restrictions relating to the exercise of the privileges of the respective license or certificate are specified.⁸¹

License is defined as the document issued by ANAC that formalizes the certification of a person to act in civil air operations, based on the fulfillment of age requirements, level of education, psychophysical aptitude, theoretical knowledge, flight instruction, experience, and proficiency, verified in accordance with the functions, limitations, and prerogatives pertaining to said license.⁸²

2. Types of License

Section 61.5 (a) of RBAC No. 61 states that under the terms of RBAC No. 61, the following licenses are granted for the performance of pilot duties: (1) student pilot; (2) private pilot; (3) commercial pilot; (4) multi-crew pilot; (5) airline pilot; (6) glider pilot; and (7) free balloon pilot.⁸³

3. Types of Qualification

According to Section 61.5 (b), the following qualifications are registered in the licenses indicated in Section 61.5 (a) of RBAC No. 61: (1) category qualifications; (2) class qualifications; (3) type qualifications; and (4) qualifications related to operations.⁸⁴

Section 61.5 (b)(3) lists the qualifications for the types of aircrafts that are noted on the pilots' licenses. Item (v) includes qualification for "any other aircraft," whenever deemed necessary by ANAC,⁸⁵ which appears to be the case for RPAs.

⁷⁹ ANAC RBAC-E No. 94.

⁸⁰ ANAC RBAC No. 67 Section 67.3(a).

⁸¹ ANAC RBAC No. 61 Subpart A, Section 61.2(11).

⁸² Id. Section 61.2(13).

⁸³ Id. Section 61.5 (a).

⁸⁴ Id. Section 61.5 (b).

⁸⁵ Id. Section 61.5 (b)(3)(v).

4. *Application for Licenses, Certificates and Authorizations*

The application for the granting of a license/certificate and/or an authorization in accordance with RBAC No. 61 must be done by filling out a specific form, presented to ANAC, which must be sent by electronic means made available by ANAC, after the applicant has met the requirements of age, level of education, psychophysical aptitude, theoretical knowledge, flight instruction, flight experience, and passing a proficiency exam provided for in RBAC No. 61, corresponding to the license/certificate and/or qualification required.⁸⁶

The theoretical knowledge requirement is met by passing a theoretical exam offered by ANAC or approved by ANAC, involving matters pertaining to the license or qualification required.⁸⁷ In cases where the completion of a theoretical course at a certified institution is a requirement for obtaining the license or qualification, the candidate will only be able to take the theoretical exam after having completed the theoretical course successfully.⁸⁸

5. *Validity*

An applicant for a license or certificate that meets all the requirements of RBAC No. 61, including passing the proficiency exam, will be entitled to the issuance of a provisional license/certificate valid for a period of 90 days.⁸⁹ The provisional license/certificate will be automatically revoked if any irregularity is found.⁹⁰

The pilot license and the aerospots pilot certificate (*Certificado de Piloto Aerodesportivo*, CPA⁹¹) are permanent, with the exception of the student pilot license, which loses its validity when the student pilot leaves the civil aviation teaching institution in which he was enrolled or after 24 months have elapsed from the date of the granting of its concession.⁹²

The validity of the qualifications recorded on pilot licenses or certificates for type qualification is 12 months, counted from the month in which the pilot passes the proficiency exam, with the exception of the provisions of Section 61.33(a), Subpart A, of RBAC No. 61 (discussed below in Section III).⁹³

⁸⁶ Id. Section 61.13(a).

⁸⁷ Id. Section 61.13(a)(1).

⁸⁸ Id.

⁸⁹ Id. Section 61.13(e).

⁹⁰ Id.

⁹¹ Id. Section 61.2(a)(7). This section defines CPA as the supporting document, with status lower than a license, which proves that the holder satisfies the requirements to operate an aerospots aircraft, as defined in RBAC No. 01, <https://perma.cc/K3X8-RFKY>, with a maximum take-off weight of up to 750 kg (1,654 lbs), except balloons and gliders, with the limitations and prerogatives established for the aforementioned certificate.

⁹² Id. Section 61.17(a).

⁹³ Id. Section 61.19(a)(2).

6. Type Qualification

Subpart K of RBAC No. 61 sets out the requirements to be met for the granting and revalidation of type qualification for pilots of airplanes, helicopters, airships, and power lift aircrafts, as well as the prerogatives and conditions for the exercise of the functions inherent to this type qualification.⁹⁴ For pilots employed by operators who have a training program approved by ANAC, the ground and flight training for the granting and revalidation of the type qualification must be made under the terms of the approved program, not applying, in this case, the mandatory training in civil aviation training centers (*Centro de Treinamento de Aviação Civil*, CTAC) provided for in subpart K of RBAC No. 61.⁹⁵

The applicant for a type qualification must comply with the following prerequisites, theoretical knowledge and ground training, and flight training:

Prerequisites - Subpart K, Section 61.213(a)(1) states that

- (i) To have passed the ANAC theoretical knowledge exam for airline pilots, as established in section 61.137 of RBAC No. 61, except for airship (*dirigíveis*) qualifications;
- ...
- (iii) To have passed, in the 12 months prior to the proficiency exam, an ANAC theoretical exam of visual flight rules (VFR) or instrument flight rules (IFR) aeronautical regulations, as applicable to the aircraft certification.⁹⁶

Section 61.137 of RBAC No. 61 states that an applicant for an airline pilot license must have passed the ANAC theoretical exam for the airline pilot license for the category in which he or she intends to obtain a license.⁹⁷

Theoretical Knowledge and Ground Training - Subpart K, Section 61.213(a)(2), states the following:

- (i) An applicant for a type qualification must demonstrate that he has successfully completed, in the 12 months prior to the proficiency exam, ground training for the type of aircraft on which he intends to obtain the qualification;
- (ii) The ground training must be conducted in a CTAC, in accordance with a training program approved by ANAC.
- (iii) If there is not, on the date on which the candidate starts ground training, a certified or validated CTAC by ANAC to provide it, the ground training may be given by a commercial pilot (*piloto comercial*, PC) or airline pilot (*piloto de linha aérea*, PLA) licensed and qualified on the aircraft, which must endorse such training at the candidate's individual flight log (*Caderneta Individual de Voo*, CIV).⁹⁸

⁹⁴ Id. Subpart K, Section 61.211(a).

⁹⁵ Id. Section 61.211(b).

⁹⁶ Id. Section 61.213(a)(1).

⁹⁷ Id. Subpart G, Section 61.137(a).

⁹⁸ Id. Subpart K, Section 61.213(a)(2).

RBAC No. 61 defines CIV as the legal document for verifying the experience, proof, and certification of flight hours of the aerospots pilot, private pilot, commercial pilot, airline pilot, glider pilot, or free balloon pilot operating aircrafts in private air services.⁹⁹

Flight Training – Subpart K, Section 61.213(a)(3) states that

(i) An applicant for a type qualification must demonstrate that he has successfully completed, in the six months prior to the proficiency exam, flight training for the type of aircraft on which he intends to obtain the qualification;

(ii) The flight training must be conducted in a CTAC, in accordance with a training program approved by ANAC.

(A) If flight simulation training devices (*Dispositivo de Treinamento para Simulação de Voo*, FSTD) are used, such devices must be qualified or validated by ANAC;

(iii) If on the date on which the candidate starts the flight training, there is no certified or validated CTAC by ANAC to provide it, the flight training may be given by a PC or PLA licensed and qualified on the aircraft, which must endorse such training at the candidate's CIV, provided it includes, at a minimum:

(A) 20 hours of flight time for jet aircrafts and 12 hours of flight time for turboprop or conventional aircrafts;

(B) 8 hours of flight time for helicopters with a maximum take-off weight up to 9,071 kg (20,000 lbs); or

(C) 10 hours of flight time for helicopters with a maximum take-off weight above 9,071 kg (20,000 lbs).¹⁰⁰

Section 61.213(a)(4), Subpart K, of RBAC No. 61 determines that a candidate must pass a proficiency exam performed on an aircraft or FSTD qualified or validated by ANAC.¹⁰¹

Supplementary Instruction No, 61-005, Revision B, establishes acceptable procedures for conducting ground and flight training for the granting and revalidation of type qualification.¹⁰²

III. Renewal Requirements

Section 61.33(a), Subpart A, of RBAC No. 61 states that provided that the requirements applicable to the revalidation of a license are met, the proficiency relevant to this revalidation can be carried out in the period that comprises 30 days before the beginning of the due month until 30 days after the end of the due month. After the revalidation is completed, the base month of expiration for the new validity is maintained.¹⁰³ The normal operation related to a license expired for less than

⁹⁹ Id. Subpart A, Section 61.2(a)(3).

¹⁰⁰ Id. Subpart K, Section 61.213(a)(3).

¹⁰¹ Id. Subpart K, Section 61.213(a)(4)(i).

¹⁰² ANAC Instrução Suplementar, IS No. 61-005, Revisão B, <https://perma.cc/P8W8-NSNY>.

¹⁰³ ANAC RBAC No. 61, Subpart A, Section 61.33(a).

30 days is allowed.¹⁰⁴ The normal operation related to a license expired for more than 30 days is prohibited, in any situation.¹⁰⁵ Qualifications revalidated after the deadline set forth in Section 61.33(a), Subpart A, of RBAC No. 61 will have their deadlines of validity counted from the month in which the pilot passed the proficiency exam.¹⁰⁶

To revalidate a type qualification, an applicant must successfully complete, within six months prior to the proficiency exam, ground and flight training for the revalidation of the license for the type of aircraft required;¹⁰⁷ and pass a proficiency examination conducted in accordance with Section 61.213(a)(4), Subpart K, of RBAC No. 61.¹⁰⁸

The revalidation also requires that the applicant must have ground and flight training conducted in a CTAC.¹⁰⁹ If at the date the candidate starts revalidation training, there is no CTAC certified or validated by ANAC to provide it, this training may be given by a PC or PLA licensed and qualified on the aircraft. The training should, in this case, include, at least 20% of the flight hours provided for in sections 61.213(a)(3)(iii)(A), 61,213(a)(3)(iii)(B) or 61,213(a)(3)(iii)(C), as applicable.¹¹⁰

¹⁰⁴ Id. Section 61.33(b).

¹⁰⁵ Id. Section 61.33(c).

¹⁰⁶ Id. Section 61.33(d).

¹⁰⁷ Id. Subpart K, Section 61.215(a)(1).

¹⁰⁸ Id. Section 61.215(a)(2).

¹⁰⁹ Id. Section 61.215(b).

¹¹⁰ Id. Section 61.215(c).

Canada

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SUMMARY In Canada, drones are designated as remotely piloted aircraft (RPA), subjecting them to the federal laws on aeronautics. RPA are regulated by the Aeronautics Act and Part IX of the Canadian Aviation Regulations—the latter being supplemented by standards, guidelines, and circulars issued by Transport Canada. These regulations prescribe terms for lawful drone usage; among them, individuals can find licensing requirements for various types of drone operations. Corporations seeking to fly drones within Canada must satisfy the requirements for a “Basic” or “Advanced” license; these classes are distinguished by the complexity of their corresponding maneuvers, and not the commercial or recreational nature of their operations. Should commercial drone activities exceed the limitations of their license class, a Special Flight Operations Certificate can be obtained on behalf of one’s business. Each class of license, however, presents its own fees, testing requirements, and 24-month recency requirements for maintaining its validity.

I. Introduction

A. Legislative Framework

Drone regulation is a competency of the federal legislature—despite it not being an enumerated power in the Constitution Act, 1867.¹ The Supreme Court of Canada provided the basis for this conclusion in the 1952 case of *Johannesson v. Municipality of West St. Paul*,² in which it asserted federal jurisdiction over the matter of aeronautics as within the Canadian government’s duty “to make Laws for the Peace, Order, and good Government of Canada,” which are known as the POGG powers.³ The Supreme Court explained that the usage of POGG powers is justified by the national importance of the aerospace industry, as “from its inherent nature,” its effects “concern the country as a whole.”⁴

Drones, or Remotely Piloted Aircraft (RPA), are subject to the Aeronautics Act, 1985.⁵ An RPA must comply with the act’s subsidiary regulations, the Canadian Aviation Regulations (CARs).⁶

* This report was prepared with the assistance of Law Library intern Miranda Lalla.

¹ Constitution Act, 1867 (Const.), 30 & 31 Vict., c 3, § 91 (U.K.), <https://perma.cc/MY2G-MM72>.

² *Johannesson v. City of West St. Paul (Rural Municipality)*, [1952] 1 S.C.R. 292, <https://perma.cc/D3DB-NPV3>.

³ Const. § 91.

⁴ *Johannesson v. City of West St. Paul (Rural Municipality)*, [1952] 1 S.C.R. 292, ¶ 7.

⁵ Aeronautics Act, R.S.C. 1985, c A-2, <https://perma.cc/5D4X-JA8V>.

⁶ Canadian Aviation Regulations (CARs), SOR/1996-433, as amended through Mar. 6, 2022, <https://perma.cc/WHU8-9W8D>.

Part IX of the CARs prescribes the majority of requirements for lawful drone usage, enforced by the minister of Transport⁷. Corresponding penalties regarding public safety can be found in the Criminal Code, 1985.⁸ Matters of privacy infringement are addressed in both the Privacy Act, 1985,⁹ and the Personal Information Protection and Electronic Documents Act, 2000.¹⁰ It must be noted that the legislative framework imposed by the Aeronautics Act and the CARs takes effect regardless of the RPA's intended usage. In other words, the commercial or recreational nature of one's RPA usage does not play a role in the determination of pilot certificate class or usage restrictions.

B. Terminology

An RPA is defined as “a set of configurable elements consisting of a remotely piloted aircraft, its control station, the command and control links and any other system elements required during flight operation.”¹¹ Since weight plays an important factor in determining an RPA's classification, it should be noted that, despite being encompassed by its definition, the weight of the control station is not taken into account when weighing the aircraft. However, any additional components to the body of the device, such as a camera or a propeller, contribute to the calculation of its weight.

A micro remotely piloted aircraft system (mRPA), weighs less than 250 grams. These are exempt from registration and certification requirements.¹²

A small remotely piloted aircraft system (sRPA), weighs between 250 grams and 25 kilograms. It is regulated by Subpart 1 of Part IX of the CARs.¹³

An sRPA must be registered and its operator certified in accordance with the type of operation performed by the device.

C. Operations

Different licensing requirements apply depending on whether drone operations are classified as “basic”¹⁴ or “advanced.”¹⁵ For an RPA operation to qualify as “basic,” it must meet the following five conditions, in which the drone is:

⁷ Aeronautics Act § 3.

⁸ Criminal Code, R.S.C. 1985, c C-46, <https://perma.cc/6QNR-KMNM>.

⁹ Privacy Act, R.S.C. 1985, c P-21, <https://perma.cc/DD7H-9G7Y>.

¹⁰ Personal Information Protection and Electronic Documents Act, S.C. 2000, c 5, <https://perma.cc/KQL2-ETGP>.

¹¹ CARs § 101.01(1).

¹² *Find Your Category of Drone Operation*, Transport Canada, <https://perma.cc/5ND6-S6BA> (last updated Feb. 19, 2022).

¹³ *Id.*

¹⁴ CARs § 901.55.

¹⁵ CARs § 901.64.

- flown in uncontrolled airspace,¹⁶
- flown more than 30 metres (100 feet) horizontally from bystanders,
- never flown over bystanders,¹⁷
- flown more than three nautical miles from a certified airport or a military airfield, and
- flown more than one nautical mile from a certified heliport.¹⁸

If any one of these criteria are not met, an RPA owner is deemed to be conducting “advanced” operations and will be subject to a more rigorous certification process.

II. Training and Certification Process

A. Training Requirements

No minimum flight hours are required for the allocation of either a basic or advanced operations pilot certificate. However, training hours within the framework of a recurrent training program can be used to fulfill recency requirements once a pilot certificate is acquired.¹⁹

1. Recency Requirements

Holders of the basic or advanced sRPA pilot certificate are granted a 24-month period to fulfill each set of recency requirements. These requirements can take the form of

- (a) attendance at a safety seminar endorsed by Transport Canada Aviation;
- (b) completion of an RPAS [remotely piloted aircraft system] recurrent training program designed to update pilot knowledge, which includes, human factors, environmental factors, route planning, operations near aerodromes/airports, and applicable regulations, rules and procedures; or
- (c) completion of a Transport Canada endorsed self-paced study program, which is designed to update pilot knowledge in the subjects specified in paragraph (b).²⁰

Training requirements take the form of continuing education. Corporations that fail to ensure their pilots meet their 24-month requirements can be subject to penalties of up to Canadian

¹⁶ Nav Canada, *Designated Airspace Handbook 6*, <https://perma.cc/8ZGL-QY36>. Controlled airspace is defined as airspace within which Air Traffic Control service is provided.

¹⁷ *Flying Your Drone Safely and Legally*, Transport Canada, <https://perma.cc/X2YB-JPLB> (last updated Sept. 1, 2020).

¹⁸ *Find Your Category of Drone Operation*, Transport Canada, <https://perma.cc/5ND6-S6BA> (last updated Feb. 19, 2022).

¹⁹ Standard 921 - Small Remotely Piloted Aircraft in Visual Line-Of-Sight (VLOS) - Canadian Aviation Regulations (CARs), § 921.04 (b), <https://perma.cc/8725-44L2>.

²⁰ Standard 921, CARs § 921.04.

\$5,000.²¹ A person operating an RPA must carry documentation of recency requirements fulfillment. Upon request, pilots must be able to show proof of training hours.²² In addition, they must be able to produce their pilot certification upon request.²³

2. Age Requirements

Age restrictions for registration of an sRPA vary in accordance with the certificate class. Both a certificate for basic operations, formally referred to as a “Pilot Certificate—Small Remotely Piloted Aircraft (VLOS) — Basic Operations” and that for advanced operations, formally referred to as a “Pilot Certificate—Remotely Piloted Aircraft (VLOS) — Advanced Operations” require that certain basic criteria be met for issuance by the minister of Transport. These are listed in the CARs.²⁴ They require that a registrant be at least 14 years of age; a Canadian citizen or a permanent resident of Canada; a corporation incorporated under the territorial, provincial, or federal laws of Canada; or a municipal, provincial, or federal entity. However, individuals under the age of 14, although unable to register an sRPA themselves, are permitted to conduct basic operations if under the supervision of the holder of either a basic or an advanced RPA pilot certificate.²⁵ Further exacting age requirements are requested for registration of an RPA when a pilot intends to conduct advanced operations; he or she must be at least 16 years of age.²⁶ However, as with aspiring basic operations licensees, the possibility to conduct advanced operations remains open to individuals under 16 years of age, when directly supervised by the holder of an Advanced RPA pilot certificate.²⁷

B. Testing

A pilot certificate for basic operations is granted by the minister of Transport upon completion of an RPAS examination.²⁸ This examination is known as the Small RPAS exam, and it is designed to test knowledge across multiple areas, such as air traffic rules and procedures, RPAS airframes, power plants, propulsion and systems, human factors, meteorology, navigation, flight operations, and the theory of flight and radiotelephony.²⁹ The exam is completed online, through the Transport Canada Drone Management Portal, in English or French. It must be completed

²¹ Transport Canada, *Aeronautical Information Manual (TC AIM), RPA – Remotely Piloted Aircraft 451* (Mar. 24, 2022), <https://perma.cc/MBZ9-YB3N>.

²² CARs § 901.56.

²³ CARs §§ 901.55, 901.64.

²⁴ CARs § 901.04.

²⁵ CARs § 901.54(2).

²⁶ CARs § 901.64 (a).

²⁷ CARs § 901.64 (c).

²⁸ CARs § 901.55 (b).

²⁹ Transport Canada, *Knowledge Requirements for Pilots of Remotely Piloted Aircraft Systems 250 g up to and Including 25 kg, Operating Within Visual Line-of-Sight (VLOS) - TP 15263*, <https://perma.cc/3TLB-ULPL>.

without assistance; to copy or solicit aid from any other person during the examination is a direct violation of the CARs.³⁰ The passing mark is 65%, and a retake is permitted after 24 hours.³¹

A pilot certificate for advanced operations is granted by the minister of Transport upon completion of a two-fold examination. First, the Small RPAS exam must be passed with a minimum grade of 80%.³² Once completed, a reference number will be attributed to the potential licensee, and this will be used to pass to the second phase of evaluation: an in-person flight review.³³

The flight review, performed at a Drone Flight School under an affiliated reviewer,³⁴ will serve to evaluate multiple competencies. To complete a flight review successfully, individuals must demonstrate their capacity to

- (i) describe the site survey process,
- (ii) describe emergency procedures that apply to flying an RPAS, including lost-link procedures and procedures to follow in the event of a fly-away, including who to contact,
- (iii) describe the method by which to inform Transport Canada of an incident or accident,
- (iv) successfully perform pre-flight checks of their RPAS,
- (v) perform a take-off,
- (vi) demonstrate the ability to navigate around obstacles,
- (vii) demonstrate the ability to recognize distances, and
- (viii) perform a landing.³⁵

Once the flight review is passed, an application for a pilot certificate of advanced operations can be completed on the online portal.³⁶

³⁰ CARs § 901.58.

³¹ CARs § 901.59.

³² Transport Canada, *supra* note 31.

³³ *Complete a Flight Review for Drones*, Transport Canada, <https://perma.cc/X9W6-XQ4L> (last updated May 30, 2019).

³⁴ *Become a Flight Reviewer for Drones: 3. Apply for Certificate*, Transport Canada, <https://perma.cc/YR8C-H65U> (last updated June 24, 2022).

³⁵ Standard 921, CARs § 921.02(b).

³⁶ *Drone Management Portal*, Transport Canada, <https://perma.cc/98BL-Q9QW> (last updated Aug. 25, 2021).

C. Fees

Registration is only required for drones above 250 grams—in other words, sRPAs. Registration fees cost of CAD\$5.³⁷ Individuals requiring a flight operations certificate are exempt from this fee.³⁸

Fees for the flight review are set by the reviewer. Once the flight review is passed, and an individual is deemed eligible for an advanced operations certificate, this certification costs an additional CAD\$5 fee (along with the CAD\$5 registration fee, and the flight review cost).³⁹

D. Certification/License

1. *Special Flight Operations Certificate*

Additional certification, called a Special Flight Operations Certificate (SFOC), may be of interest to commercial drone pilots seeking to perform operations beyond the scope of a basic or advanced pilot certificate.

An SFOC is required if

- you are a foreign operator (that is, not a Canadian citizen, permanent resident of Canada or a corporation incorporated by or under the laws of Canada or a province), authorized to operate a drone for the same purpose in your home country;
- you want to fly at a special aviation event or an advertised event;
- you want to fly closer to a military airport;
- you want to fly your drone beyond visual line-of-sight⁴⁰;
- your drone weighs over 25 kilograms (kg);
- you want to fly your drone at higher altitudes⁴¹;
- you want to fly your drone carrying dangerous or hazardous payloads (e.g. chemicals);
- you want to fly more than five drones at the same time.⁴²

³⁷ *Registering Your Drone: Overview*, Transport Canada, <https://perma.cc/DAE8-LSFJ> (last updated April 19, 2021).

³⁸ *Id.*

³⁹ *Getting a Drone Pilot Certificate*, Transport Canada, <https://perma.cc/6LGX-4YMC> (last updated June 14, 2022).

⁴⁰ *Fly Your Drone Beyond Visual Line-of-Sight*, Transport Canada, <https://perma.cc/6U6P-8FFQ> (last updated Oct. 14, 2020).

⁴¹ CARs § 901.25 (1).

⁴² *Get Permission for Special Drone Operations*, Transport Canada, <https://perma.cc/S9PG-C5BK> (last updated Oct. 14, 2020).

If applied for on behalf of a business, an SFOC must be registered with the legal and trade name of the corporation. If the business is incorporated outside of Canada, it must already have authorization to conduct drone operations in its country of incorporation, and this approval must be proven through an attestation, included with the SFOC application.

2. Personal Information Protection and Electronic Documents Act

Businesses operating drones in Canada must also abide by the Personal Information Protection and Electronic Documents Act (PIPEDA). This requires that meaningful consent be given when handling personal information in the course of commercial activities.⁴³ This is a consideration if a drone collects pictures, video, or any other “factual or subjective information about an identifiable individual.”⁴⁴ Privacy violations may go beyond PIPEDA and lead to charges of “voyeurism, mischief, creating a nuisance, and violations of provincial or municipal laws.”⁴⁵

III. Renewal Requirements

Although pilot certificates do not expire, individuals must comply with the recency requirements discussed in Section II.A.1., above.

⁴³ *PIPEDA Fair Information Principle 3 – Consent*, Off. Privacy Comm’r Can., <https://perma.cc/EDE5-9VUM> (last updated Aug. 2020).

⁴⁴ *PIPEDA In Brief*, Off. Privacy Comm’r Can., <https://perma.cc/W7JH-X4YA> (last updated May 2019).

⁴⁵ *Privacy Guidelines for Drone Users*, Transport Canada, <https://perma.cc/75L2-Z7RN> (last updated Dec. 2, 2021).

European Union

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SUMMARY Regulation (EU) 2018/1139 updates European Union (EU) safety legislation in the field of civil aviation and includes, among other things, rules for unmanned aircraft systems (UASs), commonly referred to as drones. In 2019, the European Commission adopted an implementing regulation that governs the operation of commercial and recreational UASs and sets out requirements for their remote pilots, as well as a delegated regulation setting out technical requirements for drones. The rules are directly applicable in the EU Member States, but they may enact national rules for operating drones for subjects falling outside the scope of the regulations.

Drone operations are divided into the following three categories: open, specific, and certified. UAS open operations are further subdivided into categories A1, A2, and A3, on the basis of operational limitations, requirements for the remote pilot, and technical requirements for UAS. In addition, depending on their technical requirements, UAS are divided into five different classes and must have a Conformité Européenne marking.

Training requirements for remote pilots differ depending on the class identification and the operation category of the UAS. In general, the minimum age for remote pilots operating a UAS in the open and specific category is 16 years. Likewise, testing requirements vary in the different UAS categories and subcategories. In general, remote pilots must complete an online training course, pass an online theoretical knowledge examination, and, depending on the UAS category, pass an additional theoretical knowledge examination. Licenses are generally valid for five years and may be renewed by either retesting or by completing a refresher training. Fees differ from Member State to Member State.

I. Introduction

Regulation (EU) 2018/1139 updates European Union (EU) safety legislation in the field of civil aviation and includes, among other things, rules for unmanned aircraft systems (UASs), commonly referred to as drones.¹ UASs are defined as “any aircraft of any operating mass flown autonomously or remotely without a pilot on board.”² Remote pilots are defined as “natural person[s] responsible for safely conducting the flight of an unmanned aircraft by operating its flight controls, either manually or, when the unmanned aircraft flies automatically, by monitoring its course and remaining able to intervene and change the course at any time.”³ “Unmanned

¹ Regulation (EU) 2018/1139, 2018 O.J. (L 212) 1, <https://perma.cc/D8E5-YASU>.

² Id. art. 3, no. 30.

³ Id. art. 3, no. 31.

aircraft system operators (“UAS operators”),” on the other hand, are “any legal or natural person[s] operating or intending to operate one or more UAS.”⁴

The regulation authorizes the European Commission to adopt implementing and delegated acts;⁵ to establish detailed provisions for the operation of drones and for personnel, including remote pilots; for the design, production, and maintenance of drones; and for the personnel, including remote pilots, involved in those activities.⁶ In 2019, the European Commission adopted an implementing regulation that governs the operation of commercial and recreational UASs and sets out requirements for their remote pilots, as well as a delegated regulation setting out technical requirements for drones.⁷

The rules are directly applicable in the EU Member States and replaced existing national rules.⁸ However, Member States may enact national rules for operating drones for subjects falling outside the scope of the regulations, such as public security or protection of privacy and personal data, rules for national competent authorities, introduction of a national UAS register, the age for operating drones, or with regard to fees for registration of UAS and for the online examination of remote pilots.⁹ The rules contained in the implementing regulation started applying on December 31, 2020; however, there are some transitional rules until December 31, 2023.¹⁰

The implementing regulation and the delegated regulation set out the framework for the safe operation of UAS in the EU. They adopt a risk-based approach, and, as such, do not differentiate between commercial and non-commercial UAS activities. Instead, they consider the weight and the specifications of the UAS and the operation it is intended to conduct.

II. Categories of Drone Operations

The implementing regulation divides drone operations into the following three categories: open, specific, and certified.¹¹ Depending on the category, drone operators require an operational authorization or certification from the national authorities. No authorization is required for the open category, whereas the specific category requires an operational authorization. Drone

⁴ Consolidated Version of Commission Implementing Regulation (EU) 2019/947 (Implementing Regulation), art. 2, no. 2, 2019 O.J. (L 152) 45, <https://perma.cc/TZ4Q-CYHV>.

⁵ Consolidated Version of the Treaty on the Functioning of the European Union (TFEU), arts. 290 & 291, para. 2, 2016 O.J. (C 202) 47, <https://perma.cc/3S78-4ZVQ>.

⁶ Regulation (EU) 2018/1139, arts. 57, 58. For a definition of implementing and delegated acts, see Consolidated Version of the Treaty on the Functioning of the European Union (TFEU), arts. 290 & 291, para. 2, 2016 O.J. (C 202) 47, <https://perma.cc/3S78-4ZVQ>.

⁷ Commission Delegated Regulation (EU) 2019/945 (Delegated Regulation), 2019 O.J. (L 152) 1, <https://perma.cc/EG8T-VR4T>; Consolidated Version of Commission Implementing Regulation (EU) 2019/947 (Implementing Regulation), 2019 O.J. (L 152) 45, <https://perma.cc/TZ4Q-CYHV>.

⁸ TFEU, art. 288, para. 2.

⁹ Regulation (EU) 2018/1139, arts. 2, 56, para. 8; Implementing Regulation, art. 18.

¹⁰ Implementing Regulation, art. 23.

¹¹ Id. art. 3.

operations that fall in the certified category require the certification of the drone, the certification of the operator and, where applicable, the licensing of the remote pilot.¹²

A. Open Category

Drone operations are classified as “open” where the following requirements, among others, are met:

- the UAS has a maximum take-off mass of less than 25 kilograms (25kg, about 55 pounds),
- the remote pilot ensures that the UAS is kept at a safe distance from people and is not flown over assemblies of people,
- the remote pilot keeps the UAS in visual line of sight operation (VLOS) at all times,
- the UAS is kept within 120 meters (about 394 feet) from the closest point of the surface of the earth,
- the UAS does not carry dangerous goods and does not drop any material.¹³

UAS “open” operations are further subdivided into categories A1, A2, and A3, on the basis of operational limitations, requirements for the remote pilot, and technical requirements for UAS.¹⁴

With regard to operational limitations, subcategory A1 UAS may not overfly assemblies of people and must be conducted, when the follow-me mode is active, up to a distance of 50 meters from the remote pilot.¹⁵ Subcategory A1 only applies to C0 or C1-class UAS.¹⁶

Subcategory A2 prohibits overflying uninvolved persons, and the UAS operations must take place at a safety distance of at least 30 meters from them. However, remote pilots may reduce the safety distance to a minimum of 5 meters from uninvolved persons when they are operating a UAS with an active low speed mode function and after they have evaluated the situation with regard to the weather conditions, the performance of the unmanned aircraft, and the segregation of the overflown area.¹⁷ Subcategory A2 only applies to class C2 UAS that are operated with an active and updated direct remote identification system and geo-awareness function.¹⁸

Lastly, for subcategory A3, the UAS operation must take place in an area where the remote pilot reasonably expects that no uninvolved person will be endangered, meaning outside urban areas, be conducted within a safety distance of at least 150 meters from residential, commercial,

¹² Id.

¹³ Id. art. 4.

¹⁴ Id. annex pt. A, UAS.OPEN.010, point (1).

¹⁵ Id. annex, pt. A, UAS.OPEN.020, points (1)-(3).

¹⁶ Id. annex, pt. A, UAS.OPEN.020, point (5). The five different class identifications are explained below, see II.D.

¹⁷ Id. annex, pt. A, UAS.OPEN.030, point (1).

¹⁸ Id. annex, pt. A, UAS.OPEN.030, point (3).

industrial, or recreational areas, and be performed with a privately built UAS of less than 25kg.¹⁹ Subcategory A3 only applies to class C2, C3, or C4 UAS.²⁰

B. Specific Category

If one of the requirements for open operations is not met, the drone operation is qualified as “specific” and requires an operational authorization. The operator must submit a risk assessment together with the application. The authorization will be granted when the authorities are convinced that the risk of operation can be mitigated. An individual authorization is not required if the drone is operated in the framework of a model aircraft club or association that has received an authorization.²¹

C. Certified Category

Drone operations are classified as “certified” when they are certified in accordance with specific enumerated standards outlined in EU law and the operation involves the

- operation over assemblies of people,
- transport of people, or
- carriage of dangerous goods that may result in high risk for third parties in case of an accident.²²

Drone flights with passengers on board, such as air taxis, would be an example.²³ Furthermore, the certified category also applies to drone operations where the national authority considers that the risk cannot be adequately mitigated without certification of the drone and the operator, and the licensing of the remote pilot.²⁴

D. Class Identification

Lastly, depending on their technical requirements, UAS are divided into five different classes and must have a Conformité Européenne (CE) marking.²⁵ The class identifications are C0, C1, C2, C3,

¹⁹ Id. annex, pt. A, UAS.OPEN.040, points (1), (2), (4)(a).

²⁰ Id. annex, pt. A, UAS.OPEN.040, points (4)(c)-(e).

²¹ Id. arts. 5, 16.

²² Id. art. 6.

²³ *Certified Category - Civil Drones*, European Union Aviation Safety Agency [EASA], <https://perma.cc/6K46-ZFAE>.

²⁴ Implementing Regulation, arts. 6, 11.

²⁵ Delegated Regulation, art. 6, para. 2, art. 8, para. 2(c), art. 9, para. 2, arts. 15, 16, annex pts. 1-5, annex pt. 7, no. 4. The CE marking indicates that the product is in conformity with the applicable requirements set out in EU legislation.

and C4. The drone class identification is supplied by the manufacturer at the time of purchase and is visible on the packaging.²⁶

III. Training and Certification Process

A. Training and Age Requirements

Training requirements for remote pilots differ depending on the class identification and the operation category of the UAS. In general, the minimum age for remote pilots operating a UAS in the open and specific category is 16 years.²⁷ However, there are exceptions to this general rule, and EU Member States may also lower the minimum age for remote pilots for certain categories. If a Member State decides to lower the minimum age, those remote pilots may only operate a UAS in that Member State.²⁸

1. *No Training/No Minimum Age*

No training is required if the UAS operations fall in subcategory A1 of the open category and the UAS either has a C0 class identification label, is privately built and has a maximum weight of less than 250 grams (250g), or was purchased before December 31, 2023, without a class identification label and a maximum weight of less than 250g.²⁹ However, the UAS operator must be familiar with the manufacturer's instructions for the UAS.³⁰

There are no minimum age requirements for remote pilots if the UAS is privately built with a maximum take-off mass of less than 250g, a toy, or operated under the direct supervision of a registered remote pilot.³¹

2. *Open Category*

For operations in subcategories A1 and A3, the remote pilot must complete an online training course, but no practical training is required.³² Remote pilots in subcategory A2 of the open category, on the other hand, must complete an online training course and self-practical training.³³ All remote pilots in the open category must be familiar with the manufacturer's instructions for the UAS.³⁴ There are no mandatory flying hours. The EU Member States are authorized to lower the minimum age for operating a UAS in the open category by four years down to 12 years;

²⁶ Id. art. 16, annex pt. 7, no. 4.

²⁷ Implementing Regulation, art. 9, para. 1.

²⁸ Id. art. 9, paras. 3, 4.

²⁹ Id. art. 20, annex, pt. A, UAS.OPEN.020, point (5)(a).

³⁰ Id. annex, pt. A, UAS.OPEN.020, point (4)(a).

³¹ Id. art. 9, para. 2.

³² Id. annex, pt. A, UAS.OPEN.020, point (4)(b), UAS.OPEN.040, point (3).

³³ Id. annex, pt. A, UAS.OPEN.030, point (2).

³⁴ Id. annex, pt. A, UAS.OPEN.020, point (4)(a), UAS.OPEN.030, point (2), UAS.OPEN.040, point (3).

however, it appears that, so far, most have not taken advantage of this option.³⁵ As an exception, Sweden lowered the minimum age requirement for operation of drones in the open category to 15 years.³⁶

3. *Specific Category*

The specific category is divided into standard scenario 1 (STS-01) and standard scenario 2 (STS-02).³⁷ STS-01 involves operating the UAS within visual line of sight (VLOS) over a controlled ground area in a populated environment, whereas STS-02 involves an operation beyond visual line of sight (BVLOS) with airspace observers over a controlled ground area in a sparsely populated area. Remote pilots operating in STS-01 must complete an online training course as well as practical skill training and assessment on preflight actions, in-flight procedures, and postflight actions.³⁸ The exact areas to be covered in the practical skill training and assessment are set out in a table contained in appendix 1, chapter 1 of the annex to the implementing regulation.³⁹ They include, among other matters, operation planning, airspace considerations, and site risk assessment; UAS preflight inspection and set-up; knowledge of the basic actions to be taken in the event of an emergency situation; maintaining an effective look-out and keeping the UAS within VLOS at all times; and shutting down and securing the UAS.

Likewise, remote pilots operating a UAS in STS-02 must complete an online training course as well as practical skill training and assessment as necessary for STS-01.⁴⁰ However, the training in STS-02 includes additional subjects and areas, such as airspace scanning for BVLOS operations conducted under STS-02.⁴¹

Member States are authorized to lower the minimum age requirement for remote pilots operating in the specific category by as much as two years down to 14 years of age; however, it appears that, with the exception of Sweden, no Member States have done this so far.⁴²

4. *Certified Category*

The regulations do not contain any specific requirements for the certified category with regard to training or age. The remote pilot is required to hold a pilot license, meaning that national laws on the licensing of pilots apply.⁴³ Further detailed requirements for UAS operations in the certified

³⁵ Id. art. 9, para. 3(a). Germany and France, among other Member States, have not lowered the minimum age for this category.

³⁶ 8 kap. 15 § Luftfartsförordning (SFS 2010:770), <https://perma.cc/C7TD-SR96>.

³⁷ Implementing Regulation, annex, app. 1, chs. 1, 2.

³⁸ Id. annex, app. 1, ch. 1, UAS.STS-01.020, point (1)(e), point (2)(a).

³⁹ Id. annex, app. 1, ch. 1, amend. A, point (2), tbl. 1.

⁴⁰ Id. annex, app. 1, ch. 2, UAS.STS-02.020, point (7), point (9)(a).

⁴¹ Id. annex, app. 1, ch. 2, amend. A, point (2), tbl. 1.

⁴² Id. art. 9, para. 3(b). Sweden lowered the minimum age for operation of drones in the specific category to 15 years. See 8 kap. 15 § Luftfartsförordning (SFS 2010:770).

⁴³ Implementing Regulation, art. 6, para. 2.

category might be added in the future. The website of the European Union Aviation Safety Agency (EASA) states, “in order to allow operations in the certified category almost all the aviation regulations will need to be amended. So this will be a major task. EASA decided to conduct this activity in multiple phases. . . .”⁴⁴

B. Testing

Testing requirements vary in the different UAS categories and subcategories.

1. *Open Category*

In addition to completing an online training course, remote pilots operating UAS in the A1 and A3 subcategories must pass an online theoretical knowledge examination.⁴⁵ The examination comprises 40 multiple-choice questions distributed appropriately across the following subjects: air safety, airspace restrictions, aviation regulation, human performance limitations, operational procedures, drone general knowledge, privacy and data protection, insurance, and security.⁴⁶ At least 75% of the questions must be answered correctly in order to pass both the online training and the exam.⁴⁷ However, both may be taken several times if the score is below 75%. The examination is taken at a competent national authority that the Member States have designated for that purpose, which also issues a “proof of completion of an online theoretical knowledge examination.”⁴⁸ As an example, the competent national authority for training and the issuance of certifications in Germany is the German Federal Aviation Authority (Luftfahrt-Bundesamt, LBA).⁴⁹ In France, it is the General Directorate of Civil Aviation (Direction Générale de l’Aviation Civile, DGAC),⁵⁰ and in Sweden, the Swedish Transport Agency (Transportstyrelsen).⁵¹

Remote pilots operating UAS in the A2 subcategory must hold a certificate of remote pilot competency issued by the competent national authority or by an entity recognized by the competent authority of the Member State where the UAS operator is registered.⁵² To obtain this certificate, after completing an online training course, A2 operators must pass the same online

⁴⁴ EASA, *supra* note 23.

⁴⁵ Implementing Regulation, annex, pt. A, UAS.OPEN.020, point (4)(b), UAS.OPEN.040, point (3).

⁴⁶ *Id.*

⁴⁷ *Id.*

⁴⁸ *Id.* arts. 17, 18(c).

⁴⁹ Luftverkehrs-Ordnung [LuftVO], Oct. 29, 2015, BGBl. I at 1894, as amended, §§ 21a, para. 1 & 21b, para. 1, <https://perma.cc/A7ZC-6ZCB>.

⁵⁰ Arrêté du 18 mai 2018 relatif aux exigences applicables aux télépilotes qui utilisent des aéronefs civils circulant sans personne à bord à des fins autres que le loisir, art. 14, as amended, <https://perma.cc/E2J8-L6N2>; *Examens théoriques - ULM - IULM - Télépilote - LAPL et PPL (A) et (H) – BPL*, Ministère de la Transition écologique et de la Cohésion des territoires, Ministère de la Transition énergétique (June 29, 2022), <https://perma.cc/GJP6-ANUS>.

⁵¹ 2 § Förordning om behöriga myndigheter på den civila luftfartens område (SFS 1994:1808), <https://perma.cc/74YR-6KMX>.

⁵² Implementing Regulation, annex, pt. A, UAS.OPEN.030 point (2).

theoretical knowledge examination as subcategories A1 and A3 remote pilots.⁵³ However, after completing and declaring the completion of the self-practical training, they must pass an additional theoretical knowledge examination provided by the competent national authority or by an entity recognized by the competent authority of the Member State where the UAS operator is registered.⁵⁴ The additional theoretical knowledge examination comprises at least 30 multiple-choice questions aimed at assessing the remote pilot's knowledge of the technical and operational mitigations for ground risk, distributed appropriately across the following subjects: meteorology, UAS flight performance, technical and operational mitigations for ground risk.⁵⁵ Seventy-five percent of the questions must be answered correctly to receive the pilot certificate of competency.⁵⁶ In Germany, the additional examination must be taken at specific examination centers designated by the German Federal Aviation Authority.⁵⁷

Similarly, in France, the additional examination must be taken at DGAC examination centers.⁵⁸ The examination is done on computers at all examination centers, but one center, in the town of Montpellier, also allows candidates to take the exam on paper.⁵⁹

In Sweden, the examination for the open category must be done online on the Swedish Transport Agency website using an online Bank-ID for purposes of identification.⁶⁰

2. *Specific Category*

The specific category contains different rules for UAS operators and remote pilots. UAS operators must obtain an operational authorization from the competent authority.⁶¹ They must conduct an operational risk assessment and submit it together with the application.⁶² The risk assessment must include, among other things, information regarding the competence of the personnel for conducting the operation, including their composition, roles, responsibilities, training, and recent experience.⁶³ Obtaining an operational authorization itself does not involve any training or testing. However, the UAS operator is responsible for the UAS operation and must comply with detailed requirements; in particular, he or she must designate a remote pilot for each operation and ensure that this remote pilot complies with all applicable training and testing requirements.⁶⁴

⁵³ Id. annex, pt. A, UAS.OPEN.030 point (2)(a).

⁵⁴ Id. annex, pt. A, UAS.OPEN.030 point (2)(c).

⁵⁵ Id.

⁵⁶ Id.

⁵⁷ LuftVO, § 21e. A list of the designated centers can be found on the website of the LBA. See *Liste der vom LBA benannten Prüfstellen für Fernpiloten (PStF)*, LBA, <https://perma.cc/BLA8-6PCT>.

⁵⁸ *Examens théoriques - ULM - IULM - Télépilote - LAPL et PPL (A) et (H) - BPL*, supra note 50.

⁵⁹ Id.

⁶⁰ *Utbildning för fjärrpiloter*, Transportstyrelsen, <https://perma.cc/YA6-ST24>.

⁶¹ Implementing Regulation, art. 5, para. 1, art. 12.

⁶² Id. art. 5, para. 2.

⁶³ Id. art. 11, para. 2(e).

⁶⁴ Id. annex, pt. B, UAS.SPEC.050, point (1)(b), point (1)(d).

The operational authorization details the scope of the authorization, any specific conditions that apply, such as the required competency of the remote pilots, and other information.⁶⁵

The remote pilot operating the UAS in the specific category must comply either with the competency requirements set out in the operational authorization by the competent national authority or in the standard scenarios.⁶⁶ In STS-01 as well as STS-02, remote pilots must complete an online training course, pass the online theoretical knowledge examination as required for subcategories A1/A3 in the open category, and pass an additional theoretical knowledge examination.⁶⁷ The additional theoretical knowledge examination consists of 40 multiple-choice questions aimed at assessing the remote pilot's knowledge of the technical and operational mitigations, distributed appropriately across the following subjects: aviation regulations, human performance limitations, operational procedures, technical and operational mitigations for ground risk, UAS general knowledge, meteorology, the flight performance of the UAS, and technical and operational mitigations for air risks.⁶⁸ However, if the remote pilot already holds a certificate of remote pilot competency, the examination consists of 30 multiple-choice questions only.⁶⁹ Seventy-five percent of the questions must be answered correctly to pass.⁷⁰

3. *Certified Category*

There are no specific provisions for testing requirements for operating a UAS in the certified category. However, as mentioned, remote pilots are required to hold a pilot license, meaning that national testing requirements for pilot licenses are applicable.

C. Fees

Fees for training, testing, and certifications differ from Member State to Member State. As an example, in Germany, fees are set out in the Fee Ordinance of the State Aviation Administration (Kostenverordnung der Luftfahrtverwaltung, LuftkostV).⁷¹ In the open category, fees for issuing a document confirming that the examination has been passed are 25 euros (about US\$25.46) in the A1/A3 subcategories and €30 (about US\$30.55) in the A2 subcategory.⁷² Renewals in all three subcategories are €15 (about US\$15.72).⁷³ In the specific category, the fee for issuing a document

⁶⁵ Id. art. 12, para. 4.

⁶⁶ Id. art. 8, para. 2, annex, pt. B, UAS.SPEC.060 point (1)(d).

⁶⁷ Id. annex, app. 1, ch. 1, UAS.STS-01.020, point (2); annex, app. 1, chapter 2, UAS.STS-02.020, point (7), attachment A, point (1).

⁶⁸ Id. annex, app. 1, ch. 1, attachment A, point (1)(a).

⁶⁹ Id. annex, app. 1, ch. 1, attachment A, point (1)(b).

⁷⁰ Id. annex, app. 1, ch. 1, attachment A, point (1)(c).

⁷¹ Kostenverordnung der Luftfahrtverwaltung [LuftkostV], Feb. 14, 1984, BGBl. I at 346, as amended, <https://perma.cc/2S6Z-RW8E>.

⁷² Id. annex III, nos. 34, 37(a), (aa).

⁷³ Id. annex III, nos. 35, 37(b).

that the examination for remote pilot has been passed is €30,⁷⁴ the fee for issuing the operational authorization is €200 to €2000 (about US\$204 to US\$2,037),⁷⁵ the renewal fee for the examination for remote pilots is €15,⁷⁶ and the fee for renewal of the operational authorization is €40 to €400 (about US\$41 to US\$407).⁷⁷ There are no specific fee provisions for the certified category. The registration of UAS operators in the open and specific category costs €20 to €50 (about US\$20 to US\$51) for natural as well as legal persons.⁷⁸ In the certified category, the registration of a drone is €100 (about US\$102).⁷⁹

As another example, in France, the fee for taking a license test for all drone pilots subject to licensing is €30 per exam.⁸⁰ Unemployed job seekers, however, can take it without charge.⁸¹

In Sweden, the cost for taking a license test in the open categories A1/A3 and A2 is Swedish Kronor 130 (SEK130, about US\$12).⁸² There are no annual fees for the open category, but there is a SEK50 (about US\$5) renewal fee.⁸³ The fee for testing requirements in the specific category is SEK800 (about US\$76), followed by a SEK500 (about US\$47) annual fee and a SEK200 (about US\$19) renewal fee.⁸⁴ For operators that are operating in the specific category under articles 5.4a, 5.5, or 5.5b of the EU regulation, additional oversight fees apply. For operations that fall under article 5.4a, that fee is SEK5,100 (about US\$482), with an additional fee of SEK1,700 (about US\$161) per license.⁸⁵ For activities that fall under article 5.5, the fee is SEK5,100 (about US\$482), with an additional fee of SEK2,300 (about US\$217) per standard scenario.⁸⁶ For activities that qualify under article 5.5b of the EU regulation, an annual fee of SEK54,000 (about US\$5,106) is charged.⁸⁷

⁷⁴ Id. annex III, no. 37(a), (bb).

⁷⁵ Id. annex VI, no. 34(a).

⁷⁶ Id. annex III, no. 37(b).

⁷⁷ Id. annex VI, no. 34(b).

⁷⁸ Luftverkehrsgesetz [LuftVG], May 10, 2007, BGBl. I at 698, as amended, § 66a, <https://perma.cc/CF86-ALGR>, in conjunction with LuftKostV, annex VI, no. 41a.

⁷⁹ LuftVG, § 66b in conjunction with LuftKostV, annex VI, no. 42.

⁸⁰ *Examens théoriques - ULM - IULM - Télépilote - LAPL et PPL (A) et (H) - BPL*, supra note 50.

⁸¹ Id.

⁸² 10 kap. 3 § Transportstyrelsens föreskrifter om avgifter (konsoliderad elektronisk utgåva) (SFS 2016:105), <https://perma.cc/C6AV-3RB9>.

⁸³ Id.

⁸⁴ Id.

⁸⁵ Id. 10 kap. 11 §.

⁸⁶ Id. 10 kap. 12 §.

⁸⁷ Id. 10 kap. 13 §.

D. Certification/License

All UAS operators who operate a drone with a take-off mass of 250g or more in the open category or any mass in the specific category must be registered in their Member State. For natural persons, the Member State is where they have their residence. For legal persons, it is where they have their principal place of business.⁸⁸ A registration obligation for UAS operators also exists for drones of less than 250g if the drone is equipped with a sensor for recording personal data, such as a camera.⁸⁹ After registration, the drone operator will receive a unique electronic registration number to be affixed to each drone.⁹⁰ A drone operator may operate multiple drones under the registration number. In Germany, this registration is done online on the website of the German Federal Aviation Authority. Furthermore, for drones in the certified category, the UAS operator must obtain a type certification from the German Federal Aviation Authority.⁹¹

In France, UAS operators who operate a drone in the open category with a take-off mass of 250g or more, or with an impact energy of 80 joules or more, or which is equipped with a sensor for recording personal data, must register via an online portal called AlphaTango, which is managed by the DGAC.⁹² Operators of UAS in the certified category are subject to the same registration requirements, but must generally also obtain authorizations from the Directorate for Civil Aviation Security (Direction de la sécurité de l'aviation civile, DSAC).⁹³ Exceptions whereby certified category operators do not need authorization from the DSAC include operations in a European STS-01 or STS-02 standard scenario, and some similar national standard scenarios.⁹⁴

In Sweden, every operator in the open and specific categories must be registered with the Swedish Transport Agency, which costs SEK50 (about US\$5).⁹⁵ In addition, operators must pay an annual fee of SEK50 to remain in the registry.⁹⁶

⁸⁸ Implementing Regulation, art. 14, paras. 5, 6.

⁸⁹ Id. art. 14, para. 5.

⁹⁰ Id. art. 14, para. 6.

⁹¹ Luftverkehrs-Zulassungs-Ordnung [LuftVZO], June 19, 1964, BGBl. I at 370, as amended, § 1, para. 8a, <https://perma.cc/9NED-9ZCH>.

⁹² Ministère de la transition écologique et de la Cohésion des territoires et Ministère de la Transition énergétique, *Enregistrement comme exploitant d'UAS* (July 12, 2022), <https://perma.cc/CA6R-LTW8>; *Exploitation de drones en catégorie ouverte*, Ministère de la transition écologique et de la Cohésion des territoires et Ministère de la Transition énergétique, July 12, 2022, <https://perma.cc/WHP4-X7ZG>.

⁹³ *Exploitation de drones en catégorie spécifique*, Ministère de la transition écologique et de la Cohésion des territoires, Ministère de la Transition énergétique, July 11, 2022, <https://perma.cc/LPP9-5KH3>.

⁹⁴ Id.

⁹⁵ 10 kap. 6 § Transportstyrelsens föreskrifter om avgifter (konsoliderad elektronisk utgåva) (SFS 2016:105).

⁹⁶ Id. 10 kap. 7 §.

IV. Renewal Requirements

A. Open Category

The two different types of documents for remote pilots in the open category are both valid for five years and may be renewed by either retesting or by completing a refresher training.⁹⁷ If the renewal takes place after the certificate has expired, the remote pilot must demonstrate his or her competency by completing an online training course followed by successfully completing an online theoretical knowledge examination or by completing the requirements for a certificate of remote pilot competency.⁹⁸ If the renewal takes place within the validity period, the remote pilot may choose to either demonstrate his or her competencies or complete refresher training.⁹⁹ There are no upper age limits.

B. Specific Category

There are no specific time limits for the validity of the operational authorization for UAS operators set out in the implementing regulation. The competent authority must specify the duration of the operational authorization in the authorization itself.¹⁰⁰ However, the operational authorization remains valid only as long as the UAS operator remains compliant with the relevant requirements of the implementing regulation and with the conditions defined in the operational authorization.¹⁰¹ The operational declaration is valid for two years.¹⁰²

The remote pilot certificates in STS-01 and STS-02 are both valid for five years.¹⁰³ For a renewal after the expiration of the certificate, the remote pilot must demonstrate his or her competency by completing an online training course, passing the online theoretical knowledge examination, and passing an additional theoretical knowledge examination.¹⁰⁴ If the remote pilot certificate is renewed within the validity period, the remote pilot may either demonstrate his or her competencies or complete a refresher training addressing the theoretical knowledge subjects.¹⁰⁵

C. Certified Category

There are no specific provisions for renewals in the certified category. However, any national requirements for the renewal of pilot licenses would be applicable.

⁹⁷ Implementing Regulation, annex, pt. A, UAS.OPEN.070.

⁹⁸ Id. annex, pt. A, UAS.OPEN.070, point (3).

⁹⁹ Id. annex, pt. A, UAS.OPEN.070, point (2).

¹⁰⁰ Id. annex, pt. B, UAS.SPEC.080, point (1).

¹⁰¹ Id. annex, pt. B, UAS.SPEC.080, point (2).

¹⁰² Id. annex, pt. B, UAS.SPEC.085, point (1).

¹⁰³ Id. annex, app. 1, ch. 1, UAS.STS-01.020, point (3); ch. 2, UAS.STS-02.020 point (10).

¹⁰⁴ Id. annex, app. 1, ch. 1, UAS.STS-01.020, point (4); ch. 2, UAS.STS-02.020 point (11).

¹⁰⁵ Id. annex, app. 1, ch. 1, UAS.STS-01.020, point (3); ch. 2, UAS.STS-02.020 point (10).

India

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SUMMARY Drone pilot training, testing, and certification in India is regulated by the Drone Rules 2021. Under the rules, Unmanned Aerial Vehicles are grouped in three categories (aeroplane, rotorcraft, and hybrid Unmanned Aerial System) and five classifications based on their maximum all-up (gross) weight. In order to obtain a remote pilot certificate, an applicant must complete a training program by an authorized remote pilot training organization specified by the Directorate General of Civil Aviation. The duration of courses varies depending on the category or classification of drones, but most drone training courses are short and can be completed in five to eight days. Within seven days of successfully completing the training and passing the tests, the authorized remote pilot training organization is required to submit an application for remote pilot certificate on the government's online digital sky platform, providing details about the individual who has passed the test. That individual is issued a remote pilot certificate through the same digital sky platform.

I. Introduction

In August 2021, the Ministry of Civil Aviation published the Drone Rules, 2021,¹ which were issued under the authority of the Aircraft Act, 1934.² These rules replaced the Unmanned Aircraft System Rules, 2021.³ The rules define “drone” as an “unmanned aircraft system.”⁴ An unmanned aircraft system (UAS) is defined as an “aircraft that can operate autonomously or can be operated remotely without a pilot on board.”⁵ According to the government, drones include “Unmanned Aerial Vehicles (UAVs) or Remotely Piloted Aerial Systems (RPAS) that are controlled either by a pilot on the ground or with the help of technology.”⁶

Section 2 stipulates that these rules apply to

- (a) all persons owning or possessing, or engaged in leasing, operating, transferring or maintaining an unmanned aircraft system in India;
- (b) all unmanned aircraft systems that are registered in India; and

¹ Drone Rules, 2021, Gazette of India, pt. II, sec. 3 (Aug. 25, 2021), <https://perma.cc/B2KN-5CPP>; Drone (Amendment) Rules, 2022, Gazette of India, pt. II, sec. 3 (Feb. 11, 2022), <https://perma.cc/559F-P2Z8>.

² Aircraft Act, 1934, <https://perma.cc/WQ9E-C293>.

³ Unmanned Aircraft System Rules, 2020, Gazette of India, pt. II, sec. 3 (June 4, 2020), <https://perma.cc/CYT5-H5HK>.

⁴ Drone Rules, 2021, r. 3(1)(i).

⁵ Id. r. 3(1)(zb).

⁶ Press Release, Ministry of Civil Aviation, The Drone Rules, 2021: Realising Our Collective Vision of an Aatmanirbhar Bharat (Jan. 28, 2022), <https://perma.cc/88HL-QLAP>.

(c) all unmanned aircraft systems that are being operated for the time being, in or over India.⁷

Part II of the rules deals with classification of UAS. They are grouped under three categories and five classifications. The categories are aeroplane,⁸ rotorcraft,⁹ and hybrid unmanned aircraft system,¹⁰ with the sub-categories remotely piloted aircraft system,¹¹ model remotely piloted aircraft system,¹² and autonomous unmanned aircraft system.¹³

A UAS is classified based on the “maximum all-up [gross] weight including payload”

- (a) Nano unmanned aircraft system: weighing less than or equal to 250 grams;
- (b) Micro unmanned aircraft system: weighing more than 250 grams, but less than or equal to 2 kilograms;
- (c) Small unmanned aircraft system: weighing more than 2 kilograms, but less than or equal to 25 kilograms;
- (d) Medium unmanned aircraft system: weighing more than 25 kilograms, but less than or equal to 150 kilograms; and
- (e) Large unmanned aircraft system: weighing more than 150 kilograms.¹⁴

Part III of the rules deal with the type of certification of a UAS in terms of quality standards, testing, and safety features. On January 26, 2022, the Ministry of Civil Aviation published the Certification Scheme for Unmanned Aircraft Systems.¹⁵

⁷ Drone Rules, 2021, r. 2.

⁸ Id. r. 3(1)(c), “aeroplane” means any power-driven heavier than air aircraft machine deriving support for its lift in flight chiefly from aerodynamic reactions on surfaces which remain fixed under given conditions of flight.

⁹ Id. r. 3(1)(x), “Rotorcraft” means a heavier-than-air aircraft supported in flight by the reactions of the air on one or more power driven rotors on substantially vertical axes.

¹⁰ Id. r. 3(1)(m), “hybrid unmanned aircraft” means a heavier-than-air unmanned aircraft capable of vertical take-off and landing which depends principally on power-driven lift devices or engine thrust for the lift during the flight regimes and on non-rotating airfoil for lift during horizontal flight.

¹¹ Id. r. 3(1)(w), “remotely piloted aircraft system” means a remotely piloted aircraft, its associated remote pilot stations, the required command and control links and any other components as specified in the type design; “remotely piloted aircraft” means an unmanned aircraft that is piloted from a remote pilot station.

¹² Id. r. 3(1)(n), “model remotely piloted aircraft system” means a remotely piloted aircraft system, with all-up weight not exceeding 25 kilograms, used for educational, research, design, testing or recreational purpose only and operated within visual line of sight.

¹³ Id. r. 3(1)(zb).

¹⁴ Drone Rules, 2021, r. 5.

¹⁵ Ministry Of Civil Aviation Notification, New Delhi, Certification Scheme for Unmanned Aircraft Systems, S.O. 347(E), Gazette of India, pt. II, sec. 3, subsec. (ii) (Jan. 26, 2022), <https://perma.cc/ELP2-BFQM>.

II. Training and Certification Process

A. Training Requirements

As noted in Section I above, the procedure for obtaining a remote pilot certificate includes a requirement to complete the training specified by the Directorate General of Civil Aviation (DGCA) for that specific category, sub-category, or classification of UAS. Training is only imparted by an “authorised remote pilot training organisation,”¹⁶ and “[n]o remote pilot training organisation shall be authorised [to] impart training unless it meets with the eligibility criteria as may be specified by the Director General.”¹⁷

The duration of courses varies depending on the category or classification of drones but most drone training courses are short and can be completed in five to eight days.

The syllabus for training and examination of remote pilots requires training organizations to give remote pilot training to any individual seeking a Cat-1 visual line-of-sight (VLOS) or Cat-2 beyond visual line-of-sight (BVLOS) Remote Pilot Certificate.¹⁸

The syllabus is divided into “Ground Class” and “Flying Class.” The duration of ground class for Cat-1 is broken down as follows:

- a. Theory [Ground Class] – 2 Days
- b. Practical [Flying Class including Simulator] – 3 - 6 Days
- c. Total Cat-1 [Basic] Duration – 5 - 8 Days Approx.¹⁹

The duration of the Cat-2 syllabus is broken down as follows:

- a. Theory [Ground Class] – 106 Hours Approx.
- b. Practical [Flying Class including Simulator] – 44 Hours Approx.
- c. Total Cat 2 [BVLOS] Hours – 150 Hours Approx.²⁰

B. Testing

The procedure for obtaining a remote pilot certificate also requires passing the tests conducted by the authorized remote pilot training organization. After completing ground classes, the candidate takes a “theory test.” Passing that test leads to “flying training,” which ends with a “skill test.” A candidate who passes the skill test acquires a certificate.²¹

¹⁶ Drone Rules, 2021, r. 37.

¹⁷ Id. r. 38.

¹⁸ DGCA, *Drone Training Circular No. 02 of 2022*, E-File No. DGCA-31018/3/2022-DRONE-Dte (Feb. 15 2022), <https://perma.cc/B2E7-469N>.

¹⁹ Id. at 2.

²⁰ Id.

²¹ DGCA, *Remote Pilot License Examination & Syllabus: Category 1*, <https://perma.cc/NZT3-JG9Q>.

C. Fees

According to news reports, the fee for a drone pilot training course “ranges between ₹30,000 [about US\$37] and ₹1 lakh [about US\$1,257], depending on the course.”²² In May 2022, the minister of Civil Aviation stated that fees for drone pilot training courses would decrease within the next three to four months as more training schools and institutes are certified.

Issuance or renewal of a remote pilot certificate costs 100 rupees (about US\$1.25).²³

D. Certification/License/UAS Registration

The Drone (Amendment) Rules, 2022, were published on Feb. 11, 2022. Prior to the amendment, applicants were required to get a remote pilot certificate and a drone pilot license. The 2022 amendment has abolished the requirement to acquire a drone pilot license. The “remote pilot certificate issued by a DGCA-authorized drone school is adequate for operating drones.”²⁴

An individual is eligible to obtain a remote pilot certificate if the individual

- (a) is not less than eighteen years of age and not more than sixty-five years of age;
- (b) has passed class tenth examination or its equivalent from a recognised Board; and
- (c) has successfully completed such training as may be specified by the Director General, from any authorised remote pilot training organisation.²⁵

Part VI of the rules regulates acquiring a remote pilot certificate. Rule 31 has the general requirement that “[n]o individual other than a holder of a valid remote pilot certificate enlisted on the digital sky platform shall operate an unmanned aircraft system.”²⁶ No remote pilot certificate is required, however, for “operating a nano unmanned aircraft system” or operating a “micro unmanned aircraft system for non-commercial purposes.”²⁷

The Ministry of Civil Aviation has created the Digital Sky Platform,²⁸ which it describes as a “unique unmanned traffic management (UTM) system” that facilitates online registration and certification of drones and operators “in addition to giving instant (online) clearances to operators for every flight.” The Digital Sky Platform enables online registration of pilots, devices, and

²² Isha Sahai Bhatnagar, *Cost of Training Drone Pilots Set to Reduce in Coming Days: Scindia*, Hindustan Times (May 18, 2022), <https://perma.cc/PG2G-29B4>.

²³ Drone Rules, 2021, r. 46.

²⁴ Press Release, Ministry of Civil Aviation, Samvaad with Scindia: Minister of Civil Aviation Interacted with Farmers on Benefits and Challenges of Kisan; Drones Fees for Drone Pilot Training Course Will Decrease Within the Next 3-4 Months: Shri Scindia (May 17, 2022), <https://perma.cc/ENT5-FGMR>.

²⁵ Drone Rules, 2021, r. 33.

²⁶ Drone Rules, 2021, r. 31.

²⁷ Id. r. 36.

²⁸ *Digital Sky: Interactive Airspace Maps*, DGCA, <https://digitalsky.dgca.gov.in/home>.

service providers. Using a mobile app, an operator must request permission before operating a UAS. This is called NPNT (no permission, no take-off).²⁹

A remote pilot certificate must “specifically mention the category, sub-category and classification” of the UAS, or a combination of these, for which it is issued.

Rule 34 of the rules stipulates the procedure for obtaining a remote pilot certificate

(1) Any individual, who desires to obtain a remote pilot certificate for any category, sub-category or class of an unmanned aircraft system, or a combination thereof, shall complete the training specified by the Director General for such category, sub-category or class, and pass the tests conducted by the authorised remote pilot training organisation.³⁰

Within seven days of successful completion of the training and passing of the tests, the authorized remote pilot training organization is required to make an application for remote pilot certificate on the digital sky platform, providing details of the individual who has passed the test.³¹ That individual will be issued a remote pilot certificate through the digital sky platform.³²

Rule 14 also requires that “No person shall operate an unmanned aircraft system without first registering it on the digital sky platform and obtaining a unique identification number, unless exempted from the requirement of a unique identification number under these rules.”³³

III. Renewal Requirements

A certificate must be “renewed by the authorised remote pilot training organisation for such period as may be specified therein, subject to a maximum period of ten years.” The holder of the remote pilot certificate is required to undergo “such refresher course as may be specified by the Director General on the digital sky platform from time to time.”³⁴ As stated in Section II.A. above, persons over 65 years of age cannot obtain a remote pilot certificate.

²⁹ Jagdish Kumar, *India Opens the Sky for Drones Under the “No Permission, No Takeoff” (NPNT) System*, Com. UAV News (June 3, 2019), <https://perma.cc/8R2S-5XVH>.

³⁰ Drone Rules, 2021, r. 34(1).

³¹ Id. r. 34(2).

³² Id. r. 34(3).

³³ Id. r. 14.

³⁴ Id. r. 35.

Israel

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SUMMARY Israeli law regulates commercial drone operation, training, and certification. Licensing requirements include being of the age of majority and having appropriate training, skills, knowledge, and professional experience in the specific type of drone. License applicants must successfully pass both written and practical tests. A license for commercial drone operation is valid for 24 months and may be extended for additional 24 months upon proof that the licensee has flown an aerial vehicle (a drone, or AV) of the same AV type for a period prescribed by the regulations. Maximum age limits do not appear to apply with regard to license renewal.

I. Introduction

The Aviation Law, 5771-2011, regulates the safety and operation of aircraft in Israel.¹ The law imposes licensing requirements on the commercial operation of aircraft, including those categorized as “aerial vehicles” (drones, or AVs).²

For the purpose of regulating AVs, the Aviation Law defines “commercial operation” as the operation of AVs, such as hang gliders, powered hang gliders, paragliders, powered paragliders, and powered parachutes,³ involving the transport of passengers or the providing of services.⁴

The law authorizes the minister of Transportation and Road Safety, with the approval of the Knesset Economic Affairs Committee, to determine the types of aircraft operation that require a license and the conditions for the issuance of a license. Among the conditions prescribed by regulations are being of the minimum age and having appropriate training, skills, knowledge, and professional experience. The regulations also detail the procedures for obtaining or renewing a license and the duration of licenses.⁵

The Aviation (Aerial Vehicles) Regulations 5775-2015 (AV regulations), issued under the authority prescribed by the law, set the requirements that apply to training in, and operation of, AVs.⁶ According to the regulations, license applicants must file an application with the licensing

¹ Aviation Law, 5771-2011, SH 5771 No. 2296 p. 830, as amended, available at Nevo Legal Database, <https://perma.cc/R98H-8VD3> (in Hebrew, by subscription).

² Id. § 87(b).

³ Id. § 1.

⁴ Id. § 85.

⁵ Id. § 88.

⁶ Aviation Regulations (Aerial Vehicles) 5775-2015 (AV Regulations), KT 5775 No. 7503 p. 1090, as amended, Nevo, <https://perma.cc/HSE9-LK3Q> (in Hebrew, by subscription).

department of the Civil Aviation Authority (CAA)⁷ and attach the necessary documentation and receipt for payment of the relevant filing fees.⁸

II. Training and Certification Process

The CAA's licensing department may issue an AV operation license to qualified applicants who are 18 years or older, who have presented a valid health certificate, completed appropriate training, and exhibited "professional knowledge, experience and professional skills as required under the regulations and the type of license requested."⁹

The licensing department may refuse to grant a license if one of the following conditions exists:

- (1) The applicant gave [to the licensing department] false or incorrect information;
- (2) The applicant was convicted of an offense, which, in the opinion of the licensing department, due to its essence, severity or circumstances, [means the applicant] is not fit to hold an AV license;
- (3) The applicant operated an AV in commercial operation or training in a manner that can harm aviation safety, partly due to one of the following:
 - (a) violations of aviation law by the applicant;
 - (b) ineptitude, inadequate skills or lacking professional knowledge [as] demonstrated by the applicant.¹⁰

A. Training Requirements

A license applicant must prove successful completion of training in the type of AV and for the purpose for which the license is requested. The training's curriculum must be approved by the licensing department and given by a trainer licensed to train for the AV type for which the license is requested.¹¹

⁷ The CAA operates within the Ministry of Transportation and Road Safety under authorities prescribed by the Civil Aviation Authority Law, 5765-2005, SH 5765 No. 1980 p. 130, as amended, Nevo, <https://perma.cc/7FH4-GY6T> (in Hebrew, by subscription). The CAA's duties under the law include ensuring aviation safety, issuing licenses for the operation of airports and the manufacturing of aircraft and parts as well as enforcing aviation-related international agreements to which Israel is a party. Id. § 4.

⁸ AV Regulations § 2; Aviation Regulations (Fees for Registration, Licensing and Documentation), 5770-2009, §§ 1 & 26, KT 5770 No. 6849 p. 518, as amended, Nevo, <https://perma.cc/LCT7-K9ZV> (in Hebrew, by subscription); for relevant procedures, see *Obtaining an Unmanned Aerial Vehicle Operator's Permit*, CAA, <https://perma.cc/7P8C-7KRU> (in Hebrew).

⁹ AV Regulations § 3(a).

¹⁰ Id. § 3(b).

¹¹ Id. § 5.

1. *Commercial AV License*

Proof of proficiency in the following topics is required for granting an AV operator's commercial license:

- (a) Aviation law relating to the rights and restrictions that apply to an AV licensee;
- (b) Recognition of the air traffic provisions that apply to the operation of AVs, directives and procedures of the air traffic services, responsibilities and authority of the various urban transportation routes (UTR) units, as well as knowledge of aviation information services;
- (c) General technical knowledge regarding the AVs that these regulations apply to and detailed technical knowledge of the type of hovercraft for which the license is requested; In this regulation, "technical knowledge"—understanding the AV's properties, capabilities, performance, how it operates, its limitations, including weight limits, loading rules, operating characteristics, the behavior of the AV in various situations, various effects on the AV, detection and handling of malfunctions, and whether the tool is motorized or has an auxiliary motor—also identifying and handling malfunctions in the engine;
- (d) Principles of meteorology and the implementation of basic aviation meteorology, including air clumps, micrometeorology, wind shearing, scrambles and low-altitude thermals;
- (e) Practical navigation principles applicable to AVs as well as familiarity with navigational aids and their operation;
- (f) Safety procedures and emergency procedures in various situations, including those in a situation of hovering in stormy weather¹²

2. *AV Trainer License*

To obtain a trainer's license, the applicant must prove proficiency in all the topics listed above as required for a commercial AV operator's license. In addition, the applicant must prove proficiency in the

- (b) Principles of training in general, and the principles of AV training in particular, including: learning processes and methods, teaching aids, apprentice errors and ways of correcting them;
- (c) Human performance.¹³

B. Testing

License applicants must successfully pass both substantive and practical tests.

¹² Id. § 6(1).

¹³ Id. § 6(2).

1. Written Tests

The CAA's licensing department conducts theoretical tests to evaluate applicants' proficiency and professional knowledge in all the topics listed in Section II.A above of this report. The places, manner, dates, and minimum passing scores of the tests are published on the licensing department's website.¹⁴

Only persons who have successfully passed a training course on the required curriculum by a licensed trainer may take a theoretical test.¹⁵

The licensing department's confirmation of the successful passage of a theoretical test will be valid for a period of two years. A person may, however, review, appeal, or retake the test under conditions enumerated in the regulations.¹⁶

2. Practical Tests

An AV license applicant must demonstrate before an examiner appointed by the licensing department his or her skill in operating the relevant type of AV in these matters:

- (1) procedures, exercises and maneuvers in regular aviation and in emergency situations suitable for the type of AV for which the license is requested, its performance and limitations;
- (2) hovering in the AV during daytime only, avoiding entering the clouds, and maintaining eye contact with the ground;
- (3) use of UTR networks;
- (4) precise and smooth hovering;
- (5) application of knowledge in aviation;
- (6) the ability to make a quick decision and a way of exercising judgment;
- (7) for an applicant seeking an AV license for a motorized AV or an AV with an auxiliary motor – also the ability to maintain a pre-planned route;
- (8) For an AV trainer license applicant – also proper training capability on the AV of the type for which the license is requested.¹⁷

¹⁴ AV Regulations § 7(a)-(b).

¹⁵ Id. § 7(c).

¹⁶ Id. § 7(e)-(g).

¹⁷ Id. § 9(a).

An applicant may take the practical test only upon successful completion of the required training, passage of the written test, and accumulation of the minimum required practical experience.¹⁸

C. Practical Experience

The minimum experience required for an AV license is 250 hours flying the type of AV for which the license is requested, accumulated in the five years prior to submitting the application. Experience is proven through a personal flying log, which lists the date, hours, type of AV, and the place of takeoff and landing performed by the applicant.¹⁹

D. Medical Certification

An AV license applicant must submit a medical certificate issued by an authorized aviation doctor to the licensing department. A medical certification of fitness to operate an AV must be submitted periodically to the CAA.²⁰

E. Fees

Fees associated with training, testing, and certification are prescribed by the Aviation Regulations (Fees for Registration, Licensing and Documentation), 5770-2009 (Fees Regulations).²¹

1. Fees for Certificates of Fitness

A person applying to obtain, correct, or renew a certificate of fitness must pay the following fees, depending on the weight of the AV:

- Up to 25 kg (about 55 lb.) – Israeli Shekels (ILS) 440 (about US\$127)
- 25 kg to 150 kg (about 55 lb. to 330 lb.) - ILS1,960 (about US\$567)
- More than 150 kg to 2,000 kg (about 330 lb. to 4,409 lb.) - ILS9,140 (about US\$2,646)
- More than 2,000 kg (about 4,409 lb.) - ILS20,900 (about US\$6,050).²²

An additional amount of ILS360 (about US\$104) must be paid by applicants for an AV certificate of fitness for experimental or special use “for every hour of work of an employee of the [Aviation]

¹⁸ Id. § 9(d).

¹⁹ Id. § 8.

²⁰ For information on procedures and fees required for receipt of a medical certificate, see *Frequently Asked Questions and Answers About the Process Required to Obtain an Aviation Medical Certificate*, CAA, <https://perma.cc/4JE6-F8S3> (in Hebrew).

²¹ Aviation Regulations (Fees for Registration, Licensing and Documentation), 5770-2009, KT 5770 No. 6849 p. 518, as amended, Nevo, <https://perma.cc/LCT7-K9ZV> (in Hebrew, by subscription).

²² Id. § 11(a)(1).

Authority or of a person authorized by the Authority, which was invested in providing the service, for every hour of work beyond the first two hours”²³

2. Fees for Operating Licenses

When filing their applications, applicants for “an operating license, an air operation license or an operational operating license, or a change in the name of the license owner or an addition to the activity beyond what is prescribed in the framework of such license” must pay a fee of ILS650 (about US\$188) for an AV weighing up to 25 kg or ILS1,420 (about US\$411) if it weighs more.²⁴ AV operating license holders must also pay an annual fee of ILS1,090 (about US\$316) for an AV weighing up to 25 kg and ILS1,420 for AVs weighing less.²⁵

F. Certification

In addition to the minimum age requirement, an applicant for an AV license must submit confirmation of a clean criminal record and “proficiency in speaking and reading in Hebrew and English, at the technical level accepted in the aviation world.”²⁶

III. Renewal Requirements

An AV license is valid for 24 months and may be extended for an additional 24 months upon proof that the licensee has flown an AV of the same AV type for at least 10 hours during the preceding 12 months. An extension of an AV trainer’s license requires the licensee to prove compliance with additional training experience requirements.²⁷

There does not appear to be a maximum age limit regarding license renewal.

²³ Id. § 11(a)(2); Aviation Regulations (Procedures for Documenting Aircraft and Their Parts), 5737-1977, § 68(b)-(c), KT 5737 No. 3706 p. 1576, as amended, Nevo, <https://perma.cc/BP2N-5UKK> (in Hebrew, by subscription).

²⁴ Aviation Regulations (Fees for Registration, Licensing and Documentation), § 26(a)(9)(a)-(b).

²⁵ Id. § 26(b)(9)(a)-(b).

²⁶ For additional details, see *Obtaining an Unmanned Aerial Vehicle Operator’s Permit*, *supra* note 8.

²⁷ Aviation Regulations § 12.

Japan

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SUMMARY The Civil Aeronautics Act is the main law to regulate unmanned aircrafts. The 2021 amendment to the CAA introduced the unmanned aircraft operator licenses system. The license system will be effective in December 2022. There will be two kind of licenses: first class and second class. To obtain the license, a person must pass the physical examination, knowledge test, and practical trial test.

I. Introduction

Flights of unmanned aircrafts are mainly regulated by the Civil Aeronautics Act (CAA).¹ In addition, they are also governed by the Act Prohibiting Unmanned Aircrafts' Flights over Important Facilities and Surrounding Areas, which also regulates flight areas.² The Radio Wave Act³ requires a radio station license if the device on an unmanned aircraft uses a system for transmitting data, depending on the capacity of data transmission. Further, the operator of an unmanned aircraft with a radio device must have certain qualifications.⁴

The CAA defines unmanned aircrafts as airplanes, helicopters, gliders, airships, and other equipment specified by government ordinances that are available for aviation use, that cannot be boarded by humans due to their structure, and that can be flown by remote control or automatic operation based on programs.⁵ Since June 20, 2022, the CAA has required the registration of unmanned aircrafts which weigh 100 grams (3.53 oz)⁶ or more.⁷ They are also subject to the flight permission system.⁸ The CAA was amended in 2021 to introduce certifications of unmanned

¹ Civil Aeronautics Act (CAA), Act No. 231 of 1952, as amended by Act No. 61 of 2020.

² Act Prohibiting Unmanned Aircrafts' Flights over Important Facilities and Surrounding Areas, Act No. 9 of 2016, as amended by Act No. 61 of 2020, <https://perma.cc/78WY-L79Q> (in Japanese). The CAA also restricts flight areas.

³ Radio Wave Act, Act No. 131 of 1950, as amended by Act No. 63 of 2022.

⁴ ドローン等に用いられる無線設備について (Regarding radio device used by drone), Telecommunications Bureau of the Ministry of Internal Affairs and Communications, <https://perma.cc/2NUF-CBCB>.

⁵ CAA art. 2, item 22.

⁶ Civil Aeronautics Act Enforcement Ordinance, Ministry of Transport Ordinance No. 56 of 1952, as amended by MLIT Ordinance No. 72 of 2021 (partially not in effect yet), art. 5-2, <https://perma.cc/E7LW-DYYR> (in Japanese).

⁷ CAA art. 131-3, 131-4.

⁸ Unmanned Aircraft Registration Web Portal, MLIT, <https://perma.cc/NZK3-TF7W>.

aircrafts and licenses to operate unmanned aircrafts.⁹ This part of the 2021 amendment will take effect by December 10, 2022.¹⁰

The Ministry of Land, Infrastructure, Transport and Tourism (MLIT) is the principal regulator of aviation matters, including flights of unmanned aircrafts.¹¹

II. New License System

The license system for drone operators will be introduced in December 2022. As of July 15, 2022, the MLIT has not issued detailed regulations yet.

The 2021 CAA amendment states that the Minister of MLIT will certify the skills required to fly an unmanned aircraft in certain conditions upon application by issuing an unmanned aircraft operation license.¹² There will be two kinds of licenses: the first class license and the second-class license.¹³ These are general licenses, not commercial licenses.

Only licensed unmanned aircraft operators will be permitted to operate certified unmanned aircrafts in the following airspaces:

- (a) airspaces in which unmanned aircrafts are likely to affect the safe operation of aircraft¹⁴
- (b) airspaces which are above a densely populated area¹⁵

The MLIT ordinance specifies part (a), which includes

- airspaces above airports and their vicinity;
- airspaces designated by the MLIT to secure the safety of aircraft used by the MLIT and other administrative organs for rescue work during emergencies, and
- airspaces which are 150 meters above ground or water surface level¹⁶

In addition, only licensed unmanned aircraft operators will be permitted to operate certified unmanned aircrafts when one or more of the following operational conditions are not satisfied:

⁹ Act to Partially Amend CAA, Act No. 65 of 2021, <https://perma.cc/FZZ3-AD2D> (in Japanese).

¹⁰ Id. Sup. Provisions art. 1, item 4.

¹¹ 航空 (Aviation), MLIT, <https://perma.cc/N9YL-MSHG>.

¹² CAA, as amended by Act No. 65 of 2021, art. 132-40, 132-41 (not effective yet).

¹³ Id. art. 132-42.

¹⁴ Id. art. 132-85, para. 1, item 1 (not effective yet).

¹⁵ Id. art. 132-85, para. 1, item 2 (not effective yet).

¹⁶ Civil Aeronautics Act Enforcement Ordinance art. 236-12.

- from sunrise to sunset;
- the unmanned aircrafts and surroundings are kept within the visual line of sight of the operator;
- a certain operating distance (30 meters)¹⁷ is maintained between unmanned aircrafts and persons or properties on the ground or water surface;
- no flying over festival or event sites where many people are gathered;
- no transport of explosives, flammables, or other hazardous materials;
- no dropping of any object except for the goods specified in the relevant ordinance.¹⁸

The first class license holder will be able to operate unmanned aircrafts without taking measures to limit people's access to the area below the flight area. The second class license holder will need to take measures to limit people's access to the area below the flight area when he or she operates an unmanned aircraft in the above area and/or without one or more of the above-listed conditions.¹⁹ For such flights, a flight permit will be required before the operation.²⁰ The Minister of MLIT may add conditions to the license, such as limiting the kinds of unmanned aircrafts to be operated or adding additional operational conditions.²¹

Currently, unmanned aircraft flight not satisfying all above conditions and/or in the above areas is generally prohibited.²²

III. Training & Certification Process

The MLIT will give unmanned aircrafts operator licenses to persons who pass a physical examination, knowledge test, and practical trial test.²³ To take the tests, a fee is required.²⁴ A person who has not reached 16 years of age cannot apply for the exams.

Persons who have completed an unmanned aircraft operation course provided by one of the organizations registered with MLIT may be exempted from a part of, or from the whole of, the knowledge test or the practical trial test.²⁵ Persons who completed the unmanned operation

¹⁷ Id. art. 236-18.

¹⁸ CAA, as amended by Act No. 65 of 2021, art. 132-86, para. 2 (not effective yet).

¹⁹ Id. art. 132-42 (not effective yet).

²⁰ Id. art. 132-2, para. 3 (not effective yet).

²¹ Id. art. 132-43 (not effective yet).

²² MLIT & National Police Agency, Flight Rules on Unmanned Aircraft (UA), <https://perma.cc/NJB8-VJHX>.

²³ CAA, as amended by Act No. 65 of 2021, art. 132-47 (not effective yet).

²⁴ Id. art. 135, para. 1, item 29 (not effective yet).

²⁵ Id. art. 132-50 (not effective yet).

course and earned an unmanned operation certificate from certain private organizations will be able to utilize such experience under the new system.²⁶

The license will be valid for three years. Before renewal, the person must complete a course to learn new information that is provided by a registered organization before renewal.²⁷

The MLIT has not issued its ordinance that sets details of the test.

²⁶ ドローン「免許制」で何が変わる？ 第一人者が解説 (What will change by drone “license system”? An expert explains), ICT Business Online, <https://perma.cc/98TE-DU36>.

²⁷ CAA, as amended by Act No. 65 of 2021, art. 132-51 (not effective yet).